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Increasing acceptance of oral cryotherapy among children

- A product design project

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Öka acceptansen för oral kylterapi bland barn
- Ett produktdesignprojekt

KTH Industrial Engineering
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Sammanfattning

Målet med detta examensarbete var att genom design av en ny medicinsk produkt öka acceptans för oral kylbehandling bland barn. Produktens syfte är att kyla munnens slemhinnor för att förhindra uppkomsten av sår, så kallad oral mukosit, som är en vanlig biverkning vid vissa typer av cellgiftsbehandlingar. Den enda nu existerande formen av oral kylbehandling är en isbehandling där patienten ombeds hålla isbitar eller krossad is i munnen. Målet är att kyla munnen under hela cellgiftsbehandlingen vilken kan vara i flera timmar. Detta gör behandlingen påfrestande, speciellt för barn. Utmaningen för projektet var därför att designa en produkt som levererar likvärdig kylkapacitet som isbehandlingen men på sätt som är mindre krävande för patienten.

Designprocessen startade med observationer och intervjuer på bancanceravdelningen vid Uppsala universitetssjukhus varefter en designvision formulerades. Visionen syftar till öka acceptansen för behandlingen genom att skapa kontroll, komfort och distraktion för patienten. Parallellt med sjukhusbesöken testades olika typer av kylmetoder med enkla modeller. En iterativ process av skisser, CAD modeller, fysiska modeller, användartester och utvärdering med sjukhuspersonal ledde fram till ett slutgiltigt koncept.

Resultatet är en produkt som kyler munnen genom att kall luft sprutas in genom ett munstycke. Trycksatt medicinsk luft från sjukhusets centrala system kopplas till en kylenhet monterad på en droppställning. Beslutet att kyla munnen med luft baserades på tester där detta visade sig vara den mest komfortabla kyltekniken. En fungerande prototyp av luftkylare framställdes för att testa den valda teknikens kylkapacitet. I testet uppnåddes en genomsnittlig sänkning av muntemperatur på 11,9°C vilket kan jämföras med 13,1°C med nuvarande isbehandling.

Det centrala i slutdesignen är en ljusbaserad kontroll med vilken patienten kan styra luftflöde samt få feedback om behandlingen. Syftet med denna del är att öka acceptans för behandlingen genom att ge patienten en känsla av kontroll samt erbjuda distraktion. För att engagera föräldrarna i behandlingen innefattar även kylenheten ett grässnitt genom vilket behandlingens status kan avläsas på avstånd. Funktionsprototyper av alla delar i designen framställdes och testades på friska barn och jämfördes med isbehandling. Testet visade att sex utav nio testdeltagare föredrog luftkylning varav flera motiverade sin preferens för luftkylningen med det faktum att det fanns en patientkontroll. Testet antyder således att patientkontrollen bidrar till att ge en känsla av kontroll samt har en distraherande effekt vilket kan leda till ökad acceptans för behandlingen om lösningen skulle implementeras.

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Abstract

The aim of this thesis was to increase the acceptance of oral cryotherapy treatment among children by designing a medical device. The purpose of the device is to cool the inside of the mouth in order to prevent the incidence of ulcerations, known as mucositis, which is a common side effect for certain types of chemotherapy treatments. The only existing form of oral cryotherapy to date is an ice treatment in which patients are asked to hold ice cubes or crushed ice in their mouth. The goal is to keep the mouth cool during the whole period of the chemotherapy which sometimes last several hours, making the treatment very demanding especially for children. Thus, the challenge for the design of a new device was to provide cooling capacity equal to that of the ice treatment but in a less demanding way for the patient.

The design process started with observations and interviews at the department of pediatric oncology at Uppsala University Hospital after which a design vision was established, aimed to creating control, comfort and distraction for the patient as means of increasing the treatment acceptance. In parallel, different types of cooling methods was researched and tested using quick mock-ups. An iterative process including sketches, CAD models, physical mock-ups user tests and evaluation with medical staff at the hospital led to a final concept.

The result is a device which cools the mouth with cold air injected through a mouthpiece. Pressurized air from the hospital's central system is connected to a cooling unit mounted on an IV-pole where the air is cooled. Air cooling was chosen on the basis of being the most comfortable cooling method of those tested. A functioning cooling device was prototyped to test the capacity of air cooling. The test showed that the mean mouth temperature was lowered with 11,9 °C compared to 13,1°C of present ice treatment.

At the core of the concept is a light based control unit with which the patient can adjust the airflow and receive treatment feedback with the intention to provide control and distraction. The design of the cooling unit also contains an ambient interface to involve parents by providing simple feedback about the cooling status. All parts of the design was prototyped and tested on healthy children where it was compared to the ice treatment. The test showed that six out of nine test subjects preferred air cooling, many of whom motivated their preference by the fact that it had a patient control unit. The test suggests that the control unit is contributing to a sense of control and does provide distraction from the cooling which might lead to increased treatment acceptance if implemented.

Foreword

First and foremost we would like to thank everyone at the department of pediatric oncology at Uppsala University Hospital who chose not be mentioned by name in this report. Our sincere thanks to all the patient and parents for taking time and effort to contribute with invaluable input to this project despite going through a hard time. Thanks to Senior physician Gustaf Ljungman and PhD Tove Kamsvåg Magnusson for laying the foundation of this project with their research, as well as all the medical staff for inviting us to the clinic with open arms.

We want to thank our supervisor Teo Enlund at the department of product and service design at the Royal Institute of Technology as well as members of the Clinical innovation fellowship team Christopher Blacker, Katarina Hedbeck, Hanna Andersson and Rebecca Bejhed for supporting us throughout the project.

We also want to thank Peter Hill, Benny Sjöberg and Hans Havtun at the department of applied thermodynamics at the Royal Institute of Technology for assisting with technical expertise.

Thanks Niklas Engholm at Danfoss and Aram Ghasemi at Aircontainer AB for advice and sponsoring of cooling equipment.

Nomenclature

Chemotherapy	Cancer treatment in which anti-cancer drugs are used.
Cytostatics	Umbrella term for drugs used in chemotherapy.
Oral cryotherapy	Preventative mucositis treatment in which the oral cavity is cooled.
Ice treatment	An oral cryotherapy treatment where the mouth is cooled by ice.
Oral mucosa	The mucous membrane lining the inside of the mouth.
Oral mucositis	Inflammation and ulceration of the oral mucous membrane.
Pediatric oncology	Childhood cancer.
Treatment acceptance	Here defined as, the willingness of patients to undergo a treatment.
Treatment adherence	The extent to which a patient follows the prescribed treatment.
Hydration phase	Preparatory treatment for chemotherapy by intravenous infusion of mainly water and salts.
Child life specialist	Health care professional tasked with supporting family and patient.
Chemotherapy regimen	A regimen defining the chemotherapy drugs, doses, frequency and duration.
Incidence	The rate at which a certain medical condition occurs within a population.

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1. INTRODUCTION

This chapter begins with outlining the collaboration with external partners who have been instrumental in the development of the final result. The background and purpose of the project is also presented, followed by delimitations and design opportunities.

1.2 Collaboration

This project was done in collaboration with the pediatric oncology department at Uppsala University hospital, see figure 1, and Clinical Innovation Fellowship (CIF). Clinical Innovation Fellowship is an initiative led by Center for Technology and Health (CTMH), a collaboration between the royal institute of technology (KTH), Karoliska Institutet (KI) and Stockholm county council (SLL), with the purpose of contributing to the development and commercialization of medical technology in the Stockholm region.

Clinical innovation fellowship

A multidisciplinary CIF team consisting of engineers Katarina Hedbeck and Rebecca Bejhed, physician Christopher Blacker and industrial designer Hanna Andersson, functioned as mentors throughout the project. Their expertise combined with the ambition to turn this project into a product ready for the competitive medical market has made sure the project stayed relevant and feasible.

Uppsala University Hospital

The CIF team's strong connection to the hospital has been very fruitful for this project, allowing access to conduct observations, interviews and tests as well as receiving feedback from medical staff, patients and their parents. The collaboration with the pediatric oncology clinic has given access to information that could not have been gathered using "desktop research".

Associate professor and senior physician Gustav Ljungman and PhD Tove Kamsvåg Magnusson at the pediatric oncology department at Uppsala University Hospital has contributed with medical expertise and feedback.



Figure 1: *Uppsala University Hospital*

1.3 Background

This project was initiated by the CIF team as a result of a four month long on-site study at the Pediatric oncology department Uppsala University Hospital aimed at identifying problem areas and unmet needs at the clinic. During the study the CIF team was introduced to associate professor and senior physician Gustav Ljungman and PhD Tove Kamsvåg Magnusson and their research on oral mucositis.

Having completed a study on *Parents and children's perceptions of distress related to oral mucositis during haematopoietic stem cell transplantation* (Kamsvåg-Magnusson et al., 2014) Ljungman and Kamsvåg Magnusson initiated a multi-center study in 2015 to investigate whether cryotherapeutic treatment with ice can reduce the development of oral mucositis among children (Ljungman, G. 2015). Although ice cryotherapy has been proven to reduce the incidence of mucositis among adults, (Peterson et al., 2012) evidence is lacking when it comes to the effect on children. Another study on adults where the patient's perception of the treatment was included showed that 18% perceived ice cryotherapy as unpleasant with reasons ranging from shooting pain in the teeth to nausea (Svanberg, Birgegård and Öhrn, 2007). Although there are no studies on the acceptance of oral cryotherapy among children there is reason to believe that this might become a limiting factor when it comes to the use of this treatment.

In discussion with Ljungman the CIF team saw the potential of a novel medical device that could provide oral cooling in a more patient centered fashion with focus on maximizing acceptance. That vision later became this thesis project aimed at designing a product that will increase the acceptance of oral cryotherapy by focusing primarily on the patient experience.

1.4 Delimitations

This project entails the design of a whole new type of medical product. Due to the lack of predecessors on the market there are no mistakes to learn from and no foundation of well functioning design elements to build upon. Because of this the project are subjected to large numbers of uncertainties. A big challenge is therefore to plan, prioritize and sequence the work so that each step of the project can build on the previous one. This section describes and motivates the deliverables and delimitations of this project.

Multiple users

The goal is to include all users in the design process, these are considered to be patient, parents and nurses. The working hypothesis is that all users will affect the treatment experience for the patient and therefore the treatment acceptance. The patient will be addressed both regarding physical and cognitive needs but with emphasis on the cognitive side. The nurse will be addressed mostly from a physically practical perspective i.e. making sure the device is easy to operate and fits well with current work-flow and routines. However, it will be investigated how the design can aid the nurse in including the patient and parent in the treatment. Regarding the user-group parents, the main focus will lie in making sure they trust and feel comfortable with the treatment in an effort to prevent them from being worried and projecting their fear onto their child.

The intended main users are children between 4 and 18 years old. Younger kids have been omitted in discussion with the medical experts at the clinic in order to reduce the variation in cognitive and physical abilities within the patient group. Treatment time is estimated to 0,5-16 hours depending on the type of chemotherapy the patient is given. Thus, the device will not be designed for use during sleep.

Patient interaction

The main part of the design process revolves around how the cooling can be delivered in a way that increase the patients acceptance. Thus the main focus of this project is on the patient's interaction with the product and to design an as pleasant treatment experience as possible for the child.
All interactive design elements aimed at increasing acceptance will be prototyped as close to the intended end design as possible. This is motivated the novel nature of this project combined with the fact that the design is aimed at children, which makes it difficult to evaluate ideas without a tangible representation that can be presented to the child in tests.

Cooling

The minimum requirement for the device is that it delivers a cooling capacity at least equal to that of ice cooling. Thus, the cooling principle is evaluated on a technical basis to ensure a realistic and feasible design. However detailed mechanical design and optimization of the cooling system is not performed as it is deemed superfluous in this early stage of conceptualizing a new product. Instead the focus of the cooling principle part of the project is on finding the most comfortable means of cooling the mouth. The physical nature of this challenge makes it essential to make quick mock-ups and self-experiment with different cooling principles. Due to time constraints different cooling principles is only tested and ranked by the authors.

A functioning prototype of the cooling device is developed based on the chosen cooling principle. This is done to provide a realistic setting for testing how the interactive parts of the design contributes to treatment acceptance as well as to facilitate long term tests to compare the cooling effect with the ice treatment. Note that the engineering design of the functioning prototype is of less importance since its purpose is only to deliver cooling in the same manner as the envisioned end product. Due to ethical and regulatory reasons the cooling prototype is not tested on children suffering from mucositis, instead healthy children are recruited to a final test.

Mobility of the cooling unit is taken into consideration but not designed in detail. The device is design to allow for mobile use with the addition of external mobile power and air supply i.e. battery and air tank. Built in solutions for mobility has been left for further development.

Supporting functions

Supporting functions are defined as parts of the design that connects the product and make it work as a whole. To transfer the cooling capacity from the cooling unit to the mouth of the patient, some kind of hose and mouthpiece will be needed. Both hoses and mouthpiece are considered supporting functions and will therefore have a low priority. A temporary design and mock-ups of these parts will be made to represent the final design and to be used for tests of the patient interactive elements in the design. IV-pole clamp, cooling unit handle and hose connections are all example of supporting functions that are present in the final result but whose design have received a low priority.

Patient privacy

Due to Swedish healthcare privacy laws, photographs and written material that makes any individual patient identifiable was not permitted to be published. As a consequence, photos of patients and their family that was taken at the hospital during the research and test phase have been replaced with publicly available photos for illustrative purposes.

Due to the strict nature of regulations on medical equipment medical regulations have only been loosely taken into consideration i.e. designs resulting in apparent violation of such has been avoided.

1.5 Design opportunity

Challenges

The issue of treatment acceptance is always present at a children's hospital. A child refusing to take a shot of medicine is something most nurses see everyday. Short and demanding intervention such as administering a shot can in worst case be forced upon the patient while long demanding interventions such as surgery is often performed with local anesthesia or narcosis.

Oral cryotherapy stands out in that it is a very long treatment with relatively low intensity, and because the it takes place during chemotherapy the use of anesthesia to increase acceptance could complicate the chemotherapy treatment and is therefore much undesired. This means that without treatment acceptance the treatment simply can not be given, and even if it is accepted the quality of the treatment will be dependent on the patient's adherence to the treatment regimen. A main challenge in this project will therefore be to improve both initial acceptance to start the treatment as well as a sustained compliance to continue cooling throughout the whole treatment period.

The wide user group of children in the ages between four to eighteen also pose a challenge. Considering the difference in products and activities young children interact with and appreciate to those of a teenager, it becomes clear that finding a design that will engage both is not easy.

Considering physical comfort the oral mucosa has very sensitive tissue, and mucositis can affect almost all areas of the oral cavity. It is therefore a big challenge to provide a uniform cooling that covers all the hard to reach areas of the mouth without causing too much discomfort.

As mentioned in previous chapter, parents and nurses will also be included as users. It will require special attention to design a product that does not only fit the wide primary user group, but also the routines of the nurse and worried parents.

Opportunities

As the main challenge lies in the acceptance and adherence of oral cryotherapy treatment the focus is naturally directed towards the patient experience. There seem to be an opportunity for a device that helps the medical staff to include the patient as well as parent in the treatment. This could lead to a greater sense of control which is considered a major component of patient acceptance.

Addressing all user groups, patient, parent and nurse was deemed specifically important when designing for children. As all three groups are users of the device their individual experience will likely influence the experience of the child who heavily relies on reading subtle signals from their parents and other adults. The opportunity is therefore to design a device that does not look intimidating to the parent and integrates well with the routines of the nurse.

Oral cryotherapy is often presented as voluntary treatment and is sometimes avoided by patients who would benefit from it. By designing a dedicated device there is an opportunity to shift the perception of the treatment from an primitive invention that might be good to try to a more serious and established treatment.

It has been shown in the past within the med-tech industry that when products and treatments are designed with empathy and understanding for the patient the result holds the potential to create a domino effect. The primary opportunity for a well accepted oral cryotherapy treatment is to decrease the instances and severity of oral mucositis. If this is achieved the result will be less pain, less medication, reduced number of hospital days, increased overall treatment quality, huge monetary savings and above all happier patients. Furthermore this thesis also holds a rare opportunity of designing a medical device specifically for children.

2. FRAME OF REFERENCE

This chapter entails findings from the literature study. The first chapters 2.1. *Cancer,* 2.2 *Oral mucositis and* 2.3 *Oral cryotherapy* covers the medical part of the research. These are followed by chapter 2.4 *State of the art* where the latest product within oral cryotherapy, other types of medical cooling and medical products for children are presented. Technical frame of reference regarding refrigeration devices and Cooling techniques is presented at the end of this part.

2.1 Cancer

This chapter explains what cancer is, how it is treated and the difference between child and adulthood cancer.

2.1.1 What is cancer?

Cancer is a collection of diseases which are caused by mutations in a single or a small group of cells. The mutation damages the genes in the cells causing them to divide uncontrollably into new damaged cells.

Eventually the cancer cells can form a lump or a mass of tissue called a tumor. The primary tumor can in some cases metastasize, i.e. spread to other parts of the body creating new tumors. The leukemia cancer types does not form solid tumors, instead they affect the cells in the blood stream, (National Cancer Institute, 2015a).

Treatment

There are several treatment methods for curing or slowing down the progress of cancer. Chemotherapy, radiation therapy and surgery are the three most common. The treatments can be used individually or combined depending on the type and severity of the disease, (National Cancer Institute, 2015b).

Chemotherapy, also referred to as cytostatics, is aimed to cure or slow down the progress of cancer and is the olny treatment that can effectively treat cancer that has spread throughout the body. Usually the cytostatics are injected straight into the blood with a drip or syringe but can also be prescribed as a tablet or capsule.

During the treatment cytostatics circulates through the bloodstream, attacking cells who are in the process of dividing. Cancer cells usually divide more rapidly than healthy cells which makes them more sensitive to cytostatics. However, rapidly dividing healthy cells are also affected by the treatment which can cause more or less severe side effects. Examples of fast dividing healthy cells are mucous membrane and hair cells which is why hair loss and mucositis are common side effects from chemotherapy, (Cancerresearchuk.org, 2016).

Radiation therapy is targeted to the area where cancer cells have been detected. The treatment damages the DNA in the affected cells by radiating them with e.g. x-rays and gamma rays. Radiation therapy affects cancer cells as well as healthy cells but since the treatment is targeted to a specific part of the body, the damage is limited to cells within the irradiated area. Different parts of the body are more or less sensitive to radiation which is accounted for in the treatment plan, (National Cancer Institute, 2015c).

2.1.2 Children and cancer

Cancer among adults are in many cases caused by age or external influences. When it comes to cancer among children few types are known to be genetic but in most cases the cause is unknown. That is why some types of cancer usually connected to a certain lifestyle only affect adults, e.g. lung cancer. The incidence of different types of cancer amongst children can be correlated to the age of the child, (Björk, 2009).

Treating a child for cancer usually differs from treating an adult. In general, a child can manage higher doses of cytostatics and radiation without severe side-effects, which in some cases makes it possible to shorten the treatment time by increasing the dosage. The fact that the child is still growing is also something that has to be taken into account when planning the treatment. Growing organs are more sensitive to the side-effects and the treatment can in some cases affect the growth rate of the child, (National Cancer Institute, 2015d). Special measures are also taken on account of the low age of the patient, to make sure that the child follows normal development after the treatment, e.g. to prevent infertility, (Björk, 2009).

Nursing care is also something that differs when treating a child for cancer. The mental and social development has to be taken into account during this period of the child's life, e.g. by making sure the child can retain a normal social life and keep up with school, (Björk, 2009). Different methods are used to increase the emotional comfort of the patients during treatments.

Going through a cancer treatment is both physically and mentally demanding, often leading to a diminished quality of life. Uncontrollable factors such as long hospital visits, fear of death or fear of a specific treatment can act as stressors, provoking frustration and a perceived loss of control for both child and parent. Adjusting to those stressors rather than trying to change them can be the first step towards acceptance, creating a mental foundation from which the child and parent can build their coping strategies. Diminishing uncertainties of the treatment through education and providing support focused on building confidence can be ways of empowering the child and parent, giving them an increased sense of control and emotional comfort, (Govender et al., 2015).

A common reason for emotional discomfort during treatment is a perceived loss of control. Increasing the sense of control over the treatment which has been shown to improve the quality of life for the patient. Including the patient in decision making has been shown to decrease the stress level and increase commitment for the treatment amongst adult patients, (Jansen et al., 2004). For children going through frightful or painful treatment procedures, a perceived loss of control often cause distress and anxiety. For parents to children undergoing a cancer treatment the feeling of powerlessness often leads to distress and in many cases, being able to help during the treatment can provide a sense of control resulting in increased emotional comfort, (Hedén, 2012).

Preparation

Therapeutic medical play can be used both as a way of preparing the child for a medical procedure, and to allow the child to express their thoughts and feelings about the treatment. Education about an upcoming procedure often increase the treatment acceptance, making it less unfamiliar and intimidating.

Many pediatric oncology departments have special preparation areas with mock-ups and medical equipment used during different treatment procedures, see figure 2,3,4. These areas are used to educate patients as a way of introducing and de-dramatizing specific procedures. Both child and parent often experience an increased sense of control from being familiar with the treatment, (Burns-Nader, 2011).







Figure 2,3,4. *Equipment found in preparation rooms at Uppsala university hospital.*

Distractions

Some procedures during the cancer treatment are perceived as fearful or painful, creating anxiety and emotional discomfort for the child. For instance intravenous needle insertions to access the venous port, or the insertion of a nasogastric tube when administering enteral nutrition. Apart from the distress and anxiety experienced by the child, which in turn also affects the parent, these procedures can be both time consuming and difficult for the hospital staff administering the treatment due to low acceptance. The experiences can also affect the future acceptance and create difficulties when administering the procedure again in the future.

The perceived distress and fear centers the attention on the procedure which increase the pain perception of the child making it seem more painful than it actually is. To avoid this from happening different methods are used to divert attention from the treatment by creating a distraction.

Soap bubbles, music or watching movies on a tablet are commonly used distraction techniques during stressful procedures, see figure 5,6. Sensory input based on touch can in some cases function as a distracting element, e.g. a warm pillow to take the attention from a painful procedure, (Hedén, 2012).





Figure 5,6. *Parents and nurses cooperating in an effort to distract the child during a procedure.*

2.2 Oral Mucositis

This chapter deals with the possible side effect from chemotherapy and radiation called oral mucositis.

2.2.1 What is it?

Oral mucositis is a common side effect from radiation and chemotherapy as treatments of cancer, which causes ulceration and pain in the oral mucosa. This chapter describes the cause and symptoms of oral mucositis, who are affected as well as how it is treated.

Who is affected?

The complication occurs in some degree in a majority of high dose chemotherapies, and approximately 40% of standard dose chemotherapy treatments, (Borbasi et al., 2002). About 80% of patients receiving radiation treatment to the neck and head area will also experience oral mucositis of some degree, (Treister and Sonis, 2007). Mephalan, 5-flurouracil, Metothrexat, Etopside and Cytarabin are cytostatics with a high risk of causing mucositis. (Ljungman, 2016)

2.2.2 Treatment

Some prophylactic treatments can be used to prevent or decrease the severity of oral mucositis. For instance making sure that the patient has a good mouth hygiene before the treatment. There are also evidence that oral cryotherapy during chemotherapy works as a prophylactic treatment for oral mucositis which is administered through an ice treatment. During an ice treatment the patient uses ice or in some cases ice cream to cool the mucosa inside the mouth. The ice treatment should preferably start at the same time as the chemotherapy and continue throughout the whole treatment. Pain relief and laser treatment can be used as therapeutic treatments when the oral mucositis has already occurred, (Ljungman, 2015).

"...the condition can become so severe that patients have trouble swallowing their own saliva, these kids are sitting with a bag, spitting continuously, it's really miserable" - Senior physician Gustaf Ljungman

Cause

The fast division of cells in the oral mucosa makes them more sensitive to chemotherapy and radiation. The treatments causes direct damage to the DNA in the cells, see chapter 2.1 *Cancer*. The activation of enzymes also destroys sub mucosal cells and causes inflammation. Eventually, this leads to a breach in the protective mucosa exposing nerves and causing infections, (Treister and Sonis, 2007).

Symptoms

Typical symptoms of oral mucositis are pain, ulceration and blisters in the mouth, pain in the throat, taste perception alteration, mouth and lip dryness. A study concerning the distress related to oral mucositis during haematopoietic stem cell transplantation showed that 50% of the participants experienced severe pain in the mouth, 40% severe pain in the throat, 79% severe taste perception alteration, 51% severe ulceration in the mouth, 37% severe mouth blisters and 36% severe mouth dryness.

Among patients between the age of 10 and 13 years old oral mucositis has been reported as one of the most painful and disturbing side effects of chemotherapy. In many cases this leads to difficulties in eating, drinking, sleeping, talking and swallowing. In addition to the physical side effects, oral mucositis often affects the physical well-being of the patient and in some cases delays the ongoing treatment, (Kamsvåg-Magnusson et al., 2014).

The most affected areas of the oral mucosa, seen in figure 7, are the soft palate, lateral and ventral tongue, buccal mucosa and floor of mouth, (Ljungman, 2016).



Figure 7. *Most affected areas of the oral mucosa are highlighted in blue.*

2.3 Oral cryotherapy

Oral cryotherapy refers to a treatment that cause a local reduction in temperature in the oral mucosa (Svanberg et al., 2012). Cryotherapy, on the other hand, is a set of diverse medical therapies where the temperature of the whole, or part, of the body is lowered. The therapeutic method, temperature and effect varies greatly between different types of cryotherapeutic treatments which is why it is important to stress that oral cryotherapy should not be confused with other interventions labeled cryotherapy.

Working mechanism

Although the underlying mechanisms of oral cryotherapy have not been studied in detail the most prevailing explanation is that cooling results in reduced deposition of cytostatic drugs in the cooled area. Cooling the tissue leads to vasoconstriction i.e. contraction of the blood vessels which results in reduced blood flow. Since the cytostatic drugs used in chemotherapy is administered intravenously and spreads through the body via the bloodstream, less cytostatics are thought to reach the contracted blood vessels resulting in lowered toxic concentration and reduced mucositis (Peterson et al., 2012), see figure 8. Oral cryotherapy is therefore a prophylactic treatment, although it is also thought to have a pain relieving effect once mucositis has already been developed (Farah and Savage, 2006).

Local cryotherapy have also been used with good result to prevent other side effect of chemotherapy. Ice caps or water cooled caps are used to prevent hair loss (Grevelman, 2005) and cold mittens are used to reduce nail damage (Ishiguro et al., 2011). These interventions are thought to have the same working principle as oral cryotherapy.

"The idea is to reduce the blood flow in the mouth and thereby the effect of cytostatics in the oral mucosa" - Gustaf Ljungman, Senior physician Uppsala University Hospital

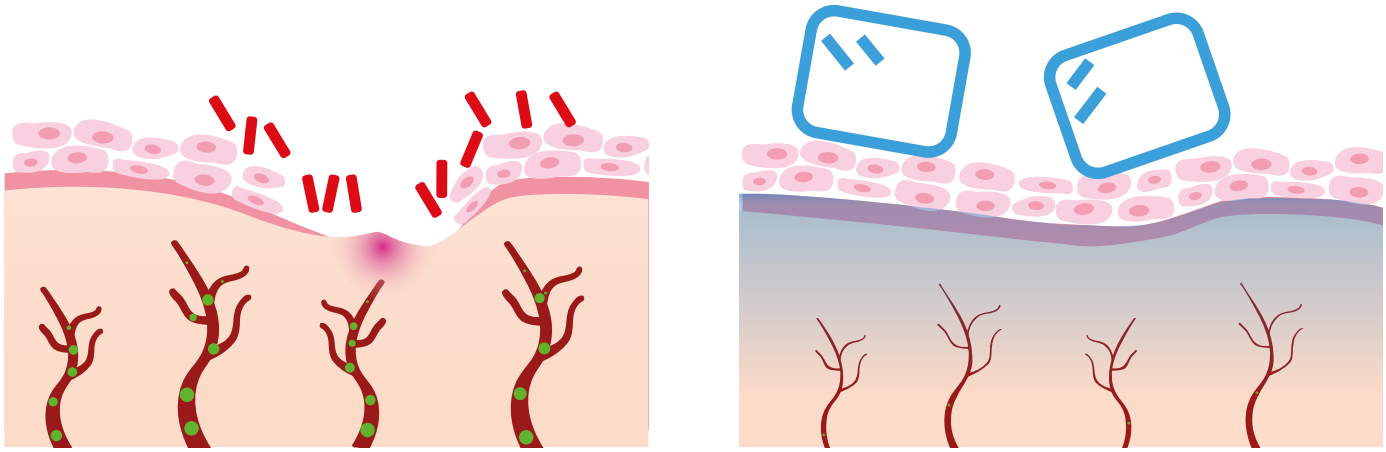


Figure 8. *The illustration depicts the working mechanism of ice treatment as seen from a cross section view of the oral mucosa. The non cooled mucosa on the right hand side is characterized by normal blood vessels that allows cytostatics, depicted as green dots, to access and damage the mucosal cells resulting in ulceration where bacteria gather to form an inflammation. The right hand side show contacted blood vessels as a result of cooling which reduce the concentration of cytostatics in the blood vessels thereby preventing damage.*

Ice treatment

Ice treatment is the only existing form of oral cryotherapy today. As the name implies the oral mucosa is cooled by ice cubes or crushed ice that the patient holds in their mouth. In some cases flavored ice or ice cream is used instead.

The ice treatment begins together with the infusion of cytostatics so that the mucosal tissue is cooled during the whole period the cytostatic drug concentration in the bloodstream is high. Treatment time could therefore range from 1-24 hours depending on the type of chemotherapy, although patients rarely cool the mouth continuously over the night.

The patient receives a container with ice and a spoon with which they fill their mouth with as much ice as they can, see figure 9. The ice starts melting immediately as it enters the mouth and the patient is instructed to constantly provide new portions as the ice is melting. The nursing staff has to refill the container with new ice throughout the duration of the treatment (Aisa et al., 2004).

Adverse effects could include headache, mouth sensitivity and nausea (National Guideline Clearinghouse, 2016). The treatment have only been researched for use on adults where the acceptance and tolerance of the treatment is often anecdotally described as "basically well tolerated" which mean that most patients were able to complete the treatment despite 39% complaining about chills and 22% had nausea (Aisa et al., 2004). One study where the participants were given a questionnaire after each cooling session showed that 18% perceived the treatment as unpleasant and out of those 10% found it very unpleasant mostly due to "shooting pain" from the teeth (Svanberg, Birgegård and Öhrn, 2007).

While ice treatment is officially recommended for some types of chemotherapy treatments in the US and used routinely at hospitals in Europe such as Uppsala University hospital there actual invention procedure of oral cryotherapy is vaguely described. The US guidelines to the treatment reads "Patients should be given small ice cubes that can be easily moved around in the mouth without causing irritation. Ice should be replenished as it melts, and patients should be instructed to move the ice in an attempt to keep the entire oral cavity cold. The use of flavored ice may be useful in promoting compliance in pediatric patients" (Anon, 2013).

With these guidelines no temperature measurement is used to determine the actual temperature in the mouth and so there is no data on what could be considered a well carried out treatment. Thus, medical staff are not able to give feedback to the patients on whether the are cooling their mouth enough, too much or too little. Neither are there any ways of knowing whether the cooling is uniform or isolated to a small part of the mouth.

The only study that has measured the temperature inside the mouth showed a 12.9°C temperature reduction after one hour of cooling with ice resulting in a mean mouth temperature of 23,8°C. This study gives a hint on how cold the mouth can be if ice is continuously added over an hour. But it does not reveal which temperature is the least required in order for the treatment to work. It should be noted that the experiment were conducted on healthy individuals with a mean age of 32 (Svanberg et al., 2012).



Figure 9. *Adult woman going through an ice treatment.*

2.4 State of the art

This chapter contains a summary of existing products related to a mouth cooler for child patients. In the first section the only patent regarding products for oral cryotherapy is presented along with the company's published plans to release a product. In the following section other cryotherapeutic devices are presented as inspiration for the many ways of cooling. Finally the last section contains medical products for children even outside cancer treatment as a compliment to the products presented in section 2.1.2 *Children and cancer*.

Cooral

Because of the recent increased in attention for mucositis the first medical device for treatment of mucositis using oral cryotherapy is expected to be released in a very near future. In may 2015 the Swedish company Braincool AB, was granted a patent on a single use oral device called Cooral, see figure 10.

Cooral consist of a hose that is constructed in a loop inside the mouth. The loose endings are to be connected to a cooling device which pumps cooled water around the loop and thus cooling the mouth (Braincool.se, 2016). Trials regarding the cooling efficiency of the product compared to ice have been performed but the results have, to the notion of the authors, not yet been published.

Although no photos has been published of the cooling unit that connects to the patented mouthpiece, Braincool AB have made the sketches presented in figure 11 available as a representation for the final product.

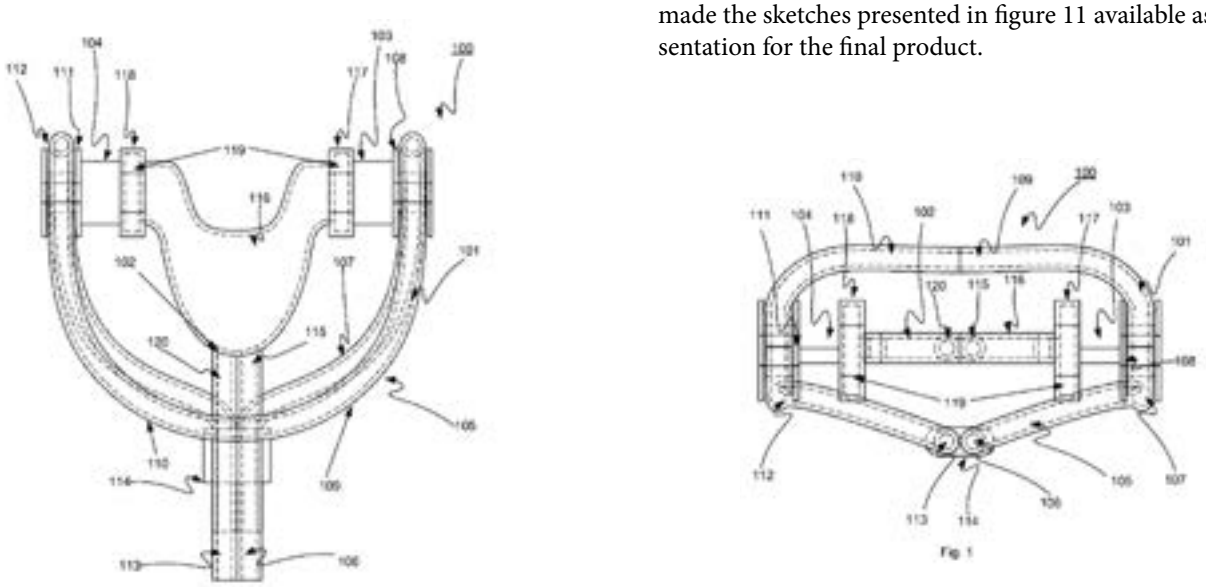


Figure 10. *Patent illustrations of the Cooral water cooled mouthpiece*



Figure 11. *Sketches of the cooling unit to which the mouthpiece is to be connected.*

2.4.1 Other cryotherapeutic products

Below follows a summary of existing products and techniques for medical cooling, see figure 12-18. This summary was done to gather inspiration on different types of cooling techniques as well as giving an idea of the size and complexity of current cryotherapeutic products. The cooling techniques and refrigeration devices used in these products are further described in the following chapters 2.5 *refrigeration technologies* and 2.6 *Cooling techniques*.



Dignicap scalp cooler

Surface cooling of the scalp by closed loop water cooling. The water is cooled by a vapor-compression refrigerator. Cooling rate is unknown (Dignicap, 2016).



Medivance Arctic sun

Closed loop water cooling lowers the temperature of the whole body by skin contact pads. The water is cooled with a vapor-compression refrigerator. Gel coated pads and negative water pressure is used to adhere the pads to the body and increase thermal conduction through the skin. Reported cooling rate is 1.5°C/hour or better (Medivance.com, 2016).



Zimmer Cryo 6

Open air cooling reduce temperature of the skin by blowing air that is cooled by a vapor-compression refrigerator. Used for reducing pain and thermal injury during laser and dermatological treatment. Reported maximum capacity of -30°C air at 1000 liters per second (Goldman, 2016).



Menopod Hot flash relief

Menopod is a device for quick relief of hot flashes caused by menopause. It is not a medical device for professional use but rather a personal treatment product. It uses direct thermoelectric cooling powered by rechargeable batteries that lasts 12 one minute cycles of cooling at 5°C (Menopod, 2016).



Zoll Thermoguard XP

Cools the whole body through a balloon catheter inserted into a femoral vein that cools the blood. The balloon catheter is cooled using a closed loop water cooling while a vapor-compression refrigerator cools the water. The product is used for precise long term cooling and have no reported cooling rate (Zoll.com, 2016).



Rhino chill

Use direct evaporative cooling by spraying a volatile and medically inert liquid into the nasal cavity and thus cool the brain. The cooling system is open and holds a small oxygen tank lasting 22 minutes and a two litre bottle of coolant that lasts one hour of cooling treatment (Benechill.com, 2016).



Emcool hypocarbon pads

Direct melt cooling with patented hypocarbon which contains a combination of graphite and water. The pads are prechilled in a freezer and attached to the skin with an adhesive layer. Reported cooling rate is up to 3.3°C/hour or 15 times more efficient than ice (Emcools.com, 2016).

Figure 12-18. *Existing cryotheraphy product.*

2.4.2 Medical products for kids

In this section a number of medical products and experiences has been summarized in order to gain inspiration on how play, storytelling or distraction might be used to increase acceptance for treatments, or just make the often serious hospital experience a little more child friendly. It should be pointed out that medical products specifically designed for children are unfortunately rare due to the relatively small markets for such products. It is often the case, as can be seen in the products shown below, products for adult are retrofitted with child friendly elements.

Super formula IV-bag cover

Ad agency JWT Brazil together with DC comics has created a hard plastic cover to fit around chemotherapy IV- bags, see figure 19. The story of a superhero formula to fight cancer is completed with accompanying comic books and animated videos featuring a superhero going through the same treatment (Jwt.com, 2013).



Figure 19. *Plastic IV-bag cover with superhero logo and accompanying magazine.*

GE Healthcare MRI scan experience room

Doug Dietz principal designer at GE Healthcare developed a child friendly MRI scan room. Each room has a theme around which a story has been built that guides the patient through the scan. The room in figure 20 below has a jungle theme and features a waterfall coming out from the wall cascading down in a coy pond under the MRI. The stream continues out into the hallway and invite kids to balance on the rocks while entering the room. Approaching the patient table, a hollowed out canoe lowers into the water allowing the child to jump into the canoe. They are told that since their lying in a canoe they have to lay still to not rock it, and if they lie very still on their back, fish will start jumping over their head, here attached on the rotating scanner head. The project has virtually eliminated the need for sedation whereas previously 80% of child patient was sedated due to anxiety. Patient satisfaction increased with 92% turning what used to be a scary experience to an exciting adventure (Kelly, 2013).



Figure 20. *MRI experience room*

Children's multi sensory unit

The multisensory unit is a cart on wheels that opens up in three sections. In figure 21 the water filled bubble tube is seen in the center, surrounded by multiple moving lights and a projector that light up the wall with colorful animations. In addition the machine also has an aromatherapy feature as well as a speaker system. Patients can control the speed of the bubbles, rhythm and the colour projected (Giving.ufhealth.org, 2013).



Figure 21. *Childrens multisensory unit*

Safety animal band

This safety animal band doubles as an elephant to make the process of needle injection less traumatic, see figure 22. It also serves the purpose of holding the needle firmly in place (Industrial Designers Society of America - IDSA, 2014).



Figure 22. *Child friendly safety band for holding the IV-needle in place.*

IV pad Lily

This skateboard like invention in figure 23 allows kids to move around in the hospital in a playful manner riding along on the Lily pad platform at the base of the IV-pole (Seeyalater.org, 2016).



Figure 23. *IV-pad Lilys with different child-friendly motives*

Video goggles

In an effort to reduce the use of sedation when performing MRI scans on child patients, Texas children's hospital has invested in video goggles, see figure 24, which also blocks the sound from the cameras as a means of distracting patients during long scans. The hospital claims the device makes the scan safer, more cost-effective and improves overall patient and parent satisfaction (Seghers, 2013).



Figure 24. *Child watching a video through video goggles.*

2.5 Refrigeration technologies

This chapter describes the technical principle behind the cooling mechanisms found in medical cooling product presented in chapter *2.4.1 Other cryotherapeutic products*. Refrigeration refers to the act of transporting heat under controlled conditions with a continuous energy input, allowing cooling below ambient temperature (Dinçer and Kanoglu, 2010). Refrigeration units can either be use in direct cooling where the cooling medium is in direct contact with what it is cooling, or indirect where a secondary medium is used to distribute the cooling capacity generated by the refrigeration unit (Granryd, 2005). For indirect cooling principles see chapter 2.6 *Cooling techniques*

Vapor-compression refrigeration

This is the most common refrigeration unit used in air conditioning and domestic fridges. A refrigerant is circulated in a closed system where it transports heat through phase change. Saturated refrigerant vapor pass through a compressor where the pressure and temperature of the gas increase, see figure 25. The hot vapor then condense in the condenser which is an air to liquid heat exchanger typically cooled by a fan blowing ambient air. The condenser is the hot side of the cycle where waste heat is discarded. After the condenser the now liquid refrigerant goes through an expansion valve where it partially turns to vapor. The refrigerant is completely vaporized in the evaporator where the phase change consumes heat energy and thus cooling the evaporator. This is the cold side of the cycle where heat is removed. After the evaporator the vaporized refrigerant enters the compressor again and the cycle is repeated (Chandra, 2016).

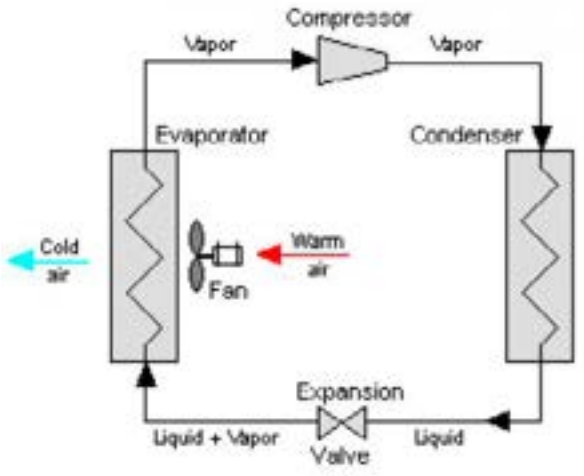


Figure 25. *Schematic illustration of a simple one stage vapor compression cycle.*

Thermoelectric refrigeration

Thermoelectric elements uses the peltier effect to convert electrical energy to a heat flux. When a direct current pass through a series of conductors, see N and P in figure 26, heat is created or removed at the junction between two conductors creating one cold and one hot side of the thermoelectric plate.

Although frequently used for cooling of electronic components the technology is far less widespread than vapor-compression refrigeration. Thermoelectric coolers holds the benefit of having no moving parts and no working liquids resulting in a compact, lightweight, maintenance free and silent refrigeration (Zhao and Tan, 2014). The main disadvantage of thermoelectric refrigeration is the coefficient of performance which is about five times lower than vapor-compression refrigeration. The low efficiency means a lot of waste heat is produced which has to be removed from the hot side in order for the cold side to remain at a low temperature. The device typically produce a maximum temperature difference of 70°C between the cold and hot side. (Enescu and Virjoghe, 2014).

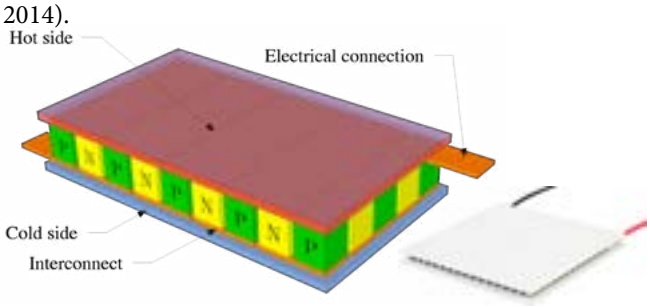


Figure 26. *Left hand side shows a schematic illustration of a peltier element, right hand side shows a photo of the same.*

Vortex tube

Vortex tube, also known as Ranque Hilsch vortex tube is a simple mechanical device with no moving parts that separates pressurised air into a cold and a hot air stream. Compressed air enters the tube in a tangential direction to the tube causing the air to swirl around creating a high speed vortex, see figure 27. A pressure difference is created between the center and the periphery of the pipe cross section. Heat energy is transported towards the high pressure peripheral section leaving a cold stream of air in the center. The hot air in the outer vortex exits the tube by the cone valve at the end of the tube while the cone force the central cold air vortex back through the tube and out on the other side (Sadi and Farzaneh-Gord, 2014). By altering the cone valve position, pressure and airspeed can be controlled to create air temperature differences of up 170°C resulting in -46 °C on the cold side and +124°C on the hot side. The device is completely driven by pressurised air typically at 5,5-6,9 bar (Exair.com, 2016).

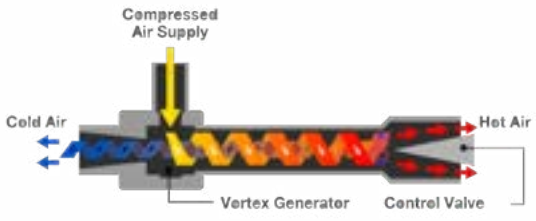


Figure 27. *Schematic illustration of a vortex tube seen from a cross sectional view*

2.6 Cooling techniques

Cooling techniques are defined in this report as indirect systems which serves the purpose of transporting distributing the cooling capacity generated in a refrigeration unit. Two of the principles described in this chapter, evaporative cooling and melt cooling, stands out in that they are not directly connected to the refrigeration unit. However they are still considered cooling principles since a refrigeration unit is required to put the cooling medium in such state where it can cool below ambient temperature through phase change e.g. a freezer is required to create ice in room temperature.
The cooling techniques presented in this chapter are represented in the products show in section 2.4.2 *Other cryotherapeutic products.*

Closed loop water cooling

In a closed loop water cooling system the water acts as the heat conductor. Using a refrigeration device, water is cooled to the desired temperature and then pumped through a closed loop system, passing through the area where the cooling is required. The water absorbs the heat from that area and is then circulated back to the refrigeration unit where it is cooled once again. Brine, a salt-water solution, is often used to minimize the risk of freezing the water in the loop as it has a lower freezing point than pure water, (Granryd, 2005). The medical cooling product presented in section 2.4.1 *Other cryotherapeutic products*, that use this cooling technique are Cooral, Dignicap scalp cooler, Medivance Arctic sun, and Zoll Thermoguard XP.

Open air cooling

Open air cooling is defined in this report as air is blown, with pre-cooled air. In order for heat transfer to appear between the surface the air is blown against and the air itself there must be a temperature difference between the two (Bergman and Incropera, 2011). To achieve cooling capacity close to that of cooling with ice or water, as in the ice treatment, the air must be cooled to below ambient room temperature. Thus the air is cooled with a refrigeration unit before it is blown onto the patient, much the same way a domestic air conditioner cools the air before blowing it into a room. This cooling technique is open meaning that the air that is blown into the mouth does not circulate back to the refrigeration unit but is wasted to the surrounding air.

When cooling room temperature air below 0°C the water vapor in the air will condense on the surface of the heat exchanger of the refrigeration unit causing frost to build up which will ultimately decrease the cooling capacity. To avoid this the air can either be dried before it is cooled or dry compressed air can be used. Some frost can be eliminated by the use of a defrosting device (Granryd, 2005). Open air cooling is used in the Zimmer cryo device presented in chapter *2.4.1 Other cryotherapeutic products*.

Evaporative cooling

Evaporative cooling occurs when a medium undergoes a phase transition from liquid to gas. The phase change is particularly important in cooling since it requires much larger amounts of heat energy, called latent heat of vaporization, than heating or cooling of the same medium within one phase. The energy needed for the phase change is drawn from the immediate surrounding of the evaporating liquid in the form of heat, causing a cooling effect (Bergman and Incropera, 2011). Evaporative cooling is the very same principle by which the human body cools itself using sweat.
The Rhino Chill intranasal cooling system presented in section 2.4.1 *Other cryotherapeutic products*, is one of few example of medical devices using evaporative cooling. It is used for cooling the brain of patients who lost blood circulation and thereby runs a risk of brain damage due to lack of oxygen. A volatile coolant, i.e. substance with low boiling point, is sprayed onto the surface of the nasal cavity at room temperature. The evaporation rate, and thereby cooling capacity, is increased by blowing oxygen from a small pressurized tank onto the wet nasal cavity surface creating forced convection (Benechill.com, 2016).

Melting phase change cooling

Melting is the transition between a medium's solid and liquid state. The energy required is called latent heat of fusion and varies between different substances where ice has a particularly high heat of fusion. The cooling effect is thus determined both by the latent heat of fusion and the melting point of the cooling substance. Besides ice, PCMs, phase change materials, is commonly used in melting phase change cooling. PCMs refers to wide variety of engineered material where the melting point is controlled. This way the temperature at which the phase change occurs can be controlled to fit a specific application (Hyperphysics.phy-astr.gsu.edu, 2016).

3. IMPLEMENTATION

The second part of this report covers the implementation of the knowledge gathered during the research phase, presented in part 2 *Frame of reference*. The design process began with learning about the users and identifying their needs. This was primarily done during visits at Uppsala university hospital and is presented in chapter 3.1 *User understanding.* The user insights were summarized and presented in chapter 3.2 *Problem definition* after which requirement specification and a design vision was established to guide the process.

The novel nature of this project led many uncertainties in the beginning of the process. It was concluded that a suitable first step was to determine a cooling technique in order to have a technically feasible foundation to build upon. Thus, this part of the report is divided in two separate design processes. The auxiliary cooling technique process, focused on finding the most comfortable means of cooling a mouth, and the main design process which was more exploratory with a stronger emphasis on the psychological needs for creating acceptance. While the two processes were carried out concurrently, the cooling principle process is presented prior to the main process in chapter 3.5 *Technical development*. The main design process begins with the following chapter 3.6 *Ideation* and continues through concept generation and user tests before the design is finalized in chapter 3.9 *Concept refinement.*

3.1 User understanding

This chapter begins with defining the different users in section 3.1.1. *Multiple users*. The first step to understand the patient is then presented in section *3.1.2 immersion* where the ice treatment was self experienced. The majority of user insights were gathered during three hospital visits. The fast changing nature of the hospital clinic made it hard to plan the research in advance. As a result unstructured research methods were used alongside semistructured interviews that were held when opportunities arose. For the sake of readability the findings from the unstructured research is described in the section *3.1.3 Unstructured hospital visit* while the semi structured interviews are presented in the section *3.1.4 User interviews*. Findings from this chapter are summarized in the following chapter 3.2 *Problem definition*.

3.1.1 Multiple users

The first part of understanding the user needs was to define the users. Three user groups were defined, patients, parents and nurses, see figure 28. The patient is the one who will undergo the treatment and was therefore naturally classified as the main user while parents and nurses were classified as secondary users.

As presented in section 2.1.2 *Children and cancer* the mood of the child is affected by the mood of the parents. The conclusions drawn from this was that treatment acceptance from child patients can only be achieved once the acceptance of parents has been earned. Parents was therefore specifically interviewed to gather insights on how they perceived various treatments and the hospital experience as a whole to detrmain how it differs from that of their child.

Since nurses will be the ones administering the treatment they were classified as secondary users. The job of a nurse already involves preparing the patient for a treatment by explaining the procedure. Insight as to how this is currently done is therefore useful to be able to design a product that makes explaining of the treatment easier. Considering that the oral cryotherapy treatment will be administered alongside chemotherapy an understanding for the current work flow of the nurse during this phase must be gained to minimize interference between the two treatments.



Figure 28. *Medical staff including and parent working together to ensure the patient feels safe.*

3.1.2 Immersion

To prepare for the hospital visits an ice treatment were simulated and self experienced by the authors, see figure 30, as a way of better understanding the treatment. The simulation consisted of constantly providing crushed ice to the mouth during a time period of fifteen minutes, which is considerably shorter than any real ice treatment.

The main conclusions from the simulation were that the treatment was very uncomfortable, on the verge to painful, especially in the beginning before getting used to the cold. The constant refilling of ice and dispose of water through swallowing or spitting added to an overall discomfort and made it hard to shift focus to anything else than the treatment.

3.1.3 Unstructured hospital visits

Unstructured interviews and observations were conducted as a way to minimize the observer influence and experience a normal day at the hospital, see figure 31. The observations were focused on the interaction between the user groups and the their feelings during a procedure rather than the procedure itself.

The interviews were held in an informal manner e.g. during rounds or in the coffee room. The staff coffee room was found to be a place where the staff spoke freely and openly in a way that made it easier to go outside the roles of designer and medical staff.

The hospital visits showed that the authors prejudice of the mood in the pediatric oncology department as a sad and gloomy place was untrue. While the severity of the child and family's situation was very tangible at times, most of the children were also seen displaying the same playfulness as one would expect from a healthy child. The children sometimes seemed to tackle the emotional distress better than their parents, seemingly more concerned about a treatment that might hurt than the overall situation.



Figure 30. *Self experiencing ice treatment.*

Another prejudice was that the often vast amount of medical equipment surrounding the child would be perceived as scary or stressful. On the contrary many of the children seemed to have an understanding for, and a relaxed approach to the surrounding equipment. One particular observation that underlined this was of a young patient who wanted to go to the play room during the hydration phase before chemotherapy. Without asking for permission the patient jumped out of the bed and pulled out the plug to the IV pump, and started leaving the room before asking her parents or nurse if it was safe.

The nurses were often observed encouraging patients to help with tasks concerning the treatment as a way of de-dramatizing it, e.g. detaching the needle from the port used as an intravenous connection.



Figure 31. *Observation during a stressful part of the day when drugs are prepared in the drug supply room.*



Figure 32. *Interview with a child life specialist in the clinic coffee room.*

Child life specialist interview

An unstructured interview was conducted with a child life specialist, see figure 32, who has a unique place in the hospital team working exclusively with helping the patient and family coping with feelings. The interview was directed towards ways of creating treatment acceptance amongst children undertaking cancer treatment.

From his experience he had learned that empowering the patient and create a feeling of responsibility can be great ways of increasing treatment acceptance. A patient controlled morphine pump, see figure 33, was brought up as a good example of a product that facilitates this. The pump allows the patient to control the dosage of morphine within a set interval . When experiencing high levels of pain the patient can press a button and receive an extra dose of morphine as opposed to calling on a nurse to adjust dosage on the pump for them. This allows the patients to control their own treatment which can create a sense of empowerment and responsibility. Furthermore the patient controlled pump, logs the number of times the patient press the button which can then be used by the medical staff to assess the experienced pain level and adjust the morphine dosage accordingly. This is particularly useful for child patients since they often have a hard time expressing pain levels accurately themselves.



Figure 33. *Patient controlled morphine pump.*

3.1.4 User Interviews

User interviews were conducted in a semi structured way. Prepared inquiries worked as an initial framework that usually led into more open discussion where the interviewee was encouraged to lead the conversation. To preserve anonymity of the participants, quotes and observations will not be referred to the name or age of the participant. Instead the participants will be referred to as patient, parent and nurse. The patient's approximate age will be classified by three different age categories as shown below.

Teenage = 13 - 18 years old.
Older child = 6 - 12 years old.
Younger child = 0 - 5 years old.

Patient

The interviews with the main user group, patients, was focused on the overall hospital experience as well as the ice treatment. Overall the ice treatment was seen as very demanding and unpleasant, see figure 34, which often lead to it being aborted or poorly implemented, especially among the young patients. Both a nurse and a patient confirmed that the treatment is worst in the beginning before getting used to the cold, something that was also experienced during the immersion phase, see section 3.1.2 *Immersion*. This finding was considered especially important since the problem occurs at the entry point to the treatment, meaning that patients might give up the whole treatment before they have overcome the first stage of numbing the mouth.

It was also found that the intensity of the cold makes it hard to concentrate on other activities normally used as distractions such as watching TV.

One patient pointed out that the treatment often is administered in the morning and therefore performed on an empty stomach, causing nausea from swallowing water from the melting ice. In addition, the patient often felt nauseous from the chemotherapy drugs even before starting the ice treatment.

Concerning the hospital experience as a whole it was confirmed that patients seem to have a curious approach to most medical equipments and that they have a need for understanding the treatments they will undergo.



Figure 34. *Selected quotes from interviews regarding the patient*

Parent

Interviews with the secondary user group, parents, focused on the role of the parent during their child's treatment and how they were affected by it.

In most cases the parent is the one the child turns to for comfort and support when being scared or sad. The mood of the parents affects the mood of the child and vice versa. Many parents felt that they have to stay strong in situations that were emotionally demanding for them, as way of protecting their child. The parent seemed to be more affected by the situation in a long term perspective as well as surroundings which acted as a reminder of the seriousness of the situation. In contrast the children seemed to be more affected by the current situation, e.g. a treatment that might hurt. This is perhaps best described by the finding that most parents seemed to have a hard time seeing nurses wearing protective clothing during the administration of chemotherapy while their children was not noticeably affected by it, see figure 35.

Parents seemed to find it hard when holding an idle role during their child's treatment, not being able to help while their child felt scared or experienced pain, something that was confirmed by the child life specialist.

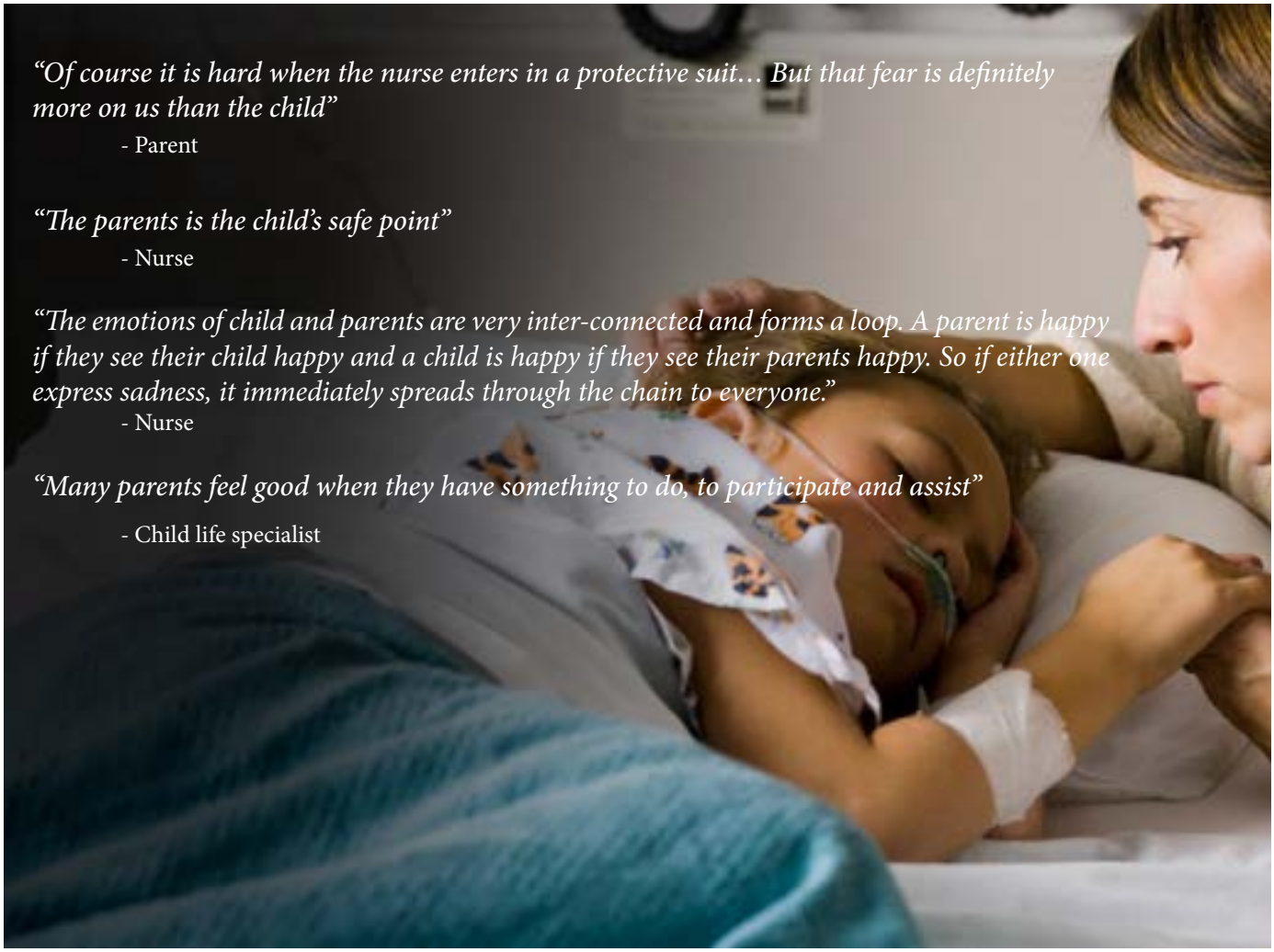


Figure 35. *Selected quotes from interviews regarding parents*

Nurse process chemotherapy

This subsection describes the process of the nurse from the moment a patient arrives at the hospital to when the chemotherapy treatment is finished. Note that surgery is only necessary if the patient suffers from leukemia.

"In the beginning, everything goes very fast in pediatric cancer care , both child and parent are in shock so you have to explain the same thing several times before they can take it in" -Nurse

"It happens that the child wakes up on the wrong side of the bed"



Figure 32

Surgery

If blood values are acceptable, patients with leukemia are sent to operation for a bone marrow sample in order to examine the response from previous treatment. If the response is not satisfactory the patient is upgraded to a new chemotherapy regimen.

Wake up

The patients that has taken a bone marrow sample wakes up from the narcosis. Sometimes this makes them more tired than usual, which makes following procedures a bit harder.

"You are so focused on getting everything right with the pump and the cytostatics that you forget to talk to the patient" -Nurse

"There are a lot of questions from the child and parents in the beginning of a treatment. But after going through the procedure a couple of times they learn the process and becomes more comfortable." -Nurse



Figure 35



Figure 36

Chemotherapy

Two nurses start by checking that the cytostatics matches the prescription. When both nurses has confirmed it is correct, the cytostatics are allowed to be administered. The responsible nurse starts by putting on protective clothing. As an extra safety procedure, the nurse checks once again that the cytostatics matches the prescription in a computer system. This is the step that requires most attention from the nurse.

Blood test

Patient usually arrives the evening before, or in the morning on the same day as the treatment. A blood sample is taken to check that blood values are sufficient to start the treatment.



Figure 33

"Most children think it's scary with needles, sometimes it can be unpleasant when accessing the vein port" -Nurse

Prehydration

Prior to cytostatics administration the patient is prehydrated intravenously with a basic solution to obtain a basic PH value. After 1-4 hours of prehydration a urine sample is taken to confirm PH is above 7.



Figure 34

"The patient is allowed to move freely during the prehydration and usually does" -Nurse

"Patients struggle in the beginning, those who continue are often motivated by the fact that they have experienced mucositis previously." -Nurse

"While undergoing the ice treatment, patients are usually sitting on their room watching TV or similar, I have never seen a patient walking around with ice in their mouth" -Nurse



Figure 37

Cryotherapy

The nurse or assisting nurse brings ice from the kitchen freezer to the patient. The patient decides if the ice should be crushed or given in whole cubes. Frozen ice cream or flavored ice can also be used on request from patient. The ice is usually delivered in a bucket and the patients feed themselves using a spoon. A glass is used for spitting and the alarm button to call on the nurse for a refill.

Post treatment

Blood tests are done every sixth hour to check concentration of cytostatics. The patient has to stay at the hospital until concentration is low. It usually takes about 24 hours. If no complications or severe adverse effects has appeared the patient can leave the hospital.

Figure 36-41. Chemotherapy process from a nurse's point of view

3.2 Problem definition

Patient

- Too cold, especially in the beginning before the tissue gets numb.
- Lack of control, patient does not know if they are cooling enough and they don't have any clear guidelines or feedback to relate to.
- Uncomfortable to swallow melting water, stomach is often empty when chemotherapy starts. Patient feels cold inside.
- Cooling takes up all attention. Hard to be distracted by other things such as tablet or TV.
- Younger children may have difficulties understanding the context of the treatment, hard to motivate why it is worth it.
- Ice has to be moved around contantly to provide even cooling.
- Hard to cool sensitive parts of the mouth, e.g soft pallet due to sensitivety and gag reflexes.
- Active treatment, patient must be motivated to constantly refill ice. This also limits other activities.
- Hard to communicate while cooling the mouth.
- Ice treatment reduces mobility.

Parents

- If the device/procedure looks intimidating from the parents point of view their fear can be projected onto the child.
- Parents often find themselves incapable of helping their child, when they become a passive bystander they tend to be more nervous.

Nurse

- The lack of temperature measurement means medical staff can not provide guidance and give correct feedback.
- Labor intensive, nurse has to refill ice bucket, patient becomes more dependent on staff.

3.3 Specification of Requirements

Demands

The device should provide a cooling effect to following parts of the mouth:

- Soft palate
- Lateral and ventral tongue
- Buccal mucosa
- Floor of mouth

The device should be capable of lowering the mean temperature of the oral mucosa by 12°C after one hour of continuous cooling.
The device should provide gradual cooling in the beginning of the treatment.
The device should be able to track the cooling progress of the oral mucosa and provide feedback on whether the cooling level is sufficient.
The patient should be able independently retrieve information about the treatment such as remaining time and cooling level.
The device should not constrain head movement while lying in bed awake.

Wishes

The cooling effect should be extended to the whole oral mucosa.
The device should provide a distracting effect for the patient, removing some of the focus of the cooling.
The device should give the patient a sense of control without compromising the treatment.
The device should allow the user to communicate orally during treatment.
The device should not constrain the patient from typical distractive activities such as watching TV, using a tablet or smart phone.
The physical restraints from the treatment should be minimized.
The treatment should be integrated in current routines and not result in a greater workload for the hospital staff than the current ice treatment.
The patient should not be forced to swallow water.
The device should not look intimidating from a parent's point of view.
The device should aid the help nurse in explain the treatment for the patient and parents.
The design should help nurse involve patient and parent in the treatment.
The design of the device should facilitate preparation.

3.4 Design vision

A design vision was constructed on the basis of the problem definition as an overall approach on how to solve the most important problems. The vision will work as a guiding document for the whole project to make sure the effort spent at each phase is on track with the fundamental vision for the project.

The diagram below, see figure 42, is a graphical representation of the design vision where three topics, control, comfort and distraction works together to create treatment acceptance.

Control

Control refers to the ability to change or affect the treatment in an intentional way. The first step towards control is establishing empowerment, which is defined as knowledge about the treatment and the cooling device. Empowerment is a vital part of control because it ensures that users not only holds the ability to affect the treatment, but that they can do so in an intentional manner. For instance a user might hold the ability to push a button to adjust the treatment, but if they are unsure of how it will change they are not in control.

It should be pointed out that the need to feel a sense of control over the treatment is not necessarily equal to the need for actually controlling it. Although actual control might be used in the beginning of a treatment it is hypothesized that once the user has established a sense of control the abillity of affecting the course of the treatment will rarely be used. Although the patient is the main focus when it comes to providing control,parents are considered almost equally important in this regard.

Another expected effect of control is that it brings responsibility which is thought to be a good motivator and thus contribute to long term treatment adherence. This builds on the idea that children are incentivised by showing parents and medical staff that they "can do it themselves".

Comfort

Comfort is thought to be connected to control in that a sense of control over the treatment allows the user to relax and therby gain phycological comfort. The physical comfort sets the objective level of what the patient must endure while undergoing the treatment and is thus a natural part in creating tratment acceptance. Naturally the design should strive for maximum physical comfort during the whole treatment but the first part, here defined as the entry point, is considered of special interest. Entry point is the point of attentional entry to the treatment and makes up the first minutes of cooling. This point is important because it sets the initial impression of the treatment which is thought to greatly influence subsequent perception and attitude. This is also the point where the current ice treatment has its biggest weakness.

Distraction

Distraction is defined in this report as is something that draws the attention away from the treatment. As presented in section 2.1.2 *Children and cancer*, different distractive techniques and product are already widely used within children's health care with good results. It is hypothesized that in order for patients to be susceptible to distractions, a certain level of comfort and control is needed. If these fundamental needs are not fulfilled a distractive element could instead become a burden or something annoying. By design, a distraction calls for the attention of the user. It is therefore an act of balance to design it in such way that it is only present when the distraction is called for. This is especially important to consider give the long treatment period of oral cryotherapy.

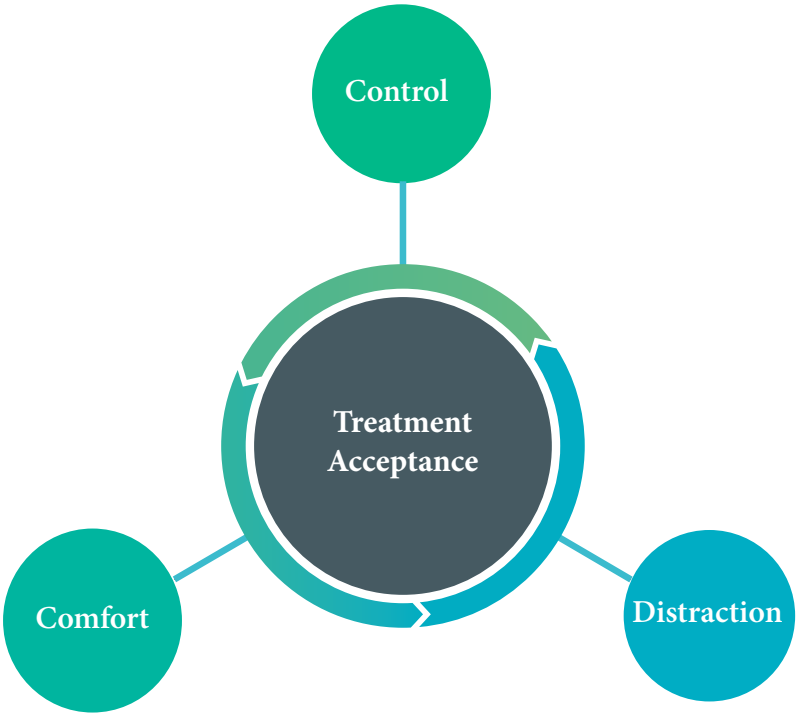


Figure 42. *Illustration of the design vision showing how the interconnected topics control, comfort and distraction is contributes to treatment acceptance.*

3.5 Technical development

The goal of this phase was to find the cooling principle best suited in terms of desired cooling capacity, comfort and practical integration in treatment routines. The chosen cooling principle functioned as a foundation for further development concerning the user experience. The process began with an ideation phase based on the initial research seen in chapter 2.4 *State of the art* and 2.6 *Cooling techniques*. The ideation phase resulted in five concepts which were evaluated and a final cooling principle determined.

3.5.1 Technical brainstorming

An interview was conducted with a research engineer, (Peter Hill), and a senior lecturer, (Hans Havtun), in thermodynamics at the Royal Institute of Technology in Stockholm. The interview was structured as a brainstorming session aimed to discuss advantages and disadvantages of the cooling principles from the initial research phase, presented in chapter 2.6 *Cooling principles*. It was also aimed to find new solutions and technologies overlooked in the initial research phase.

Both interviewees saw many advantages in open air cooling but could not confirm the technical feasibility since they lacked experience of cooling air. They advised against trying to confirm the technical feasibility with calculations and instead suggested building a prototype. This principle would not require direct contact with the oral mucosa and could make it more comfortable to cool some of the more sensitive parts of the mouth.

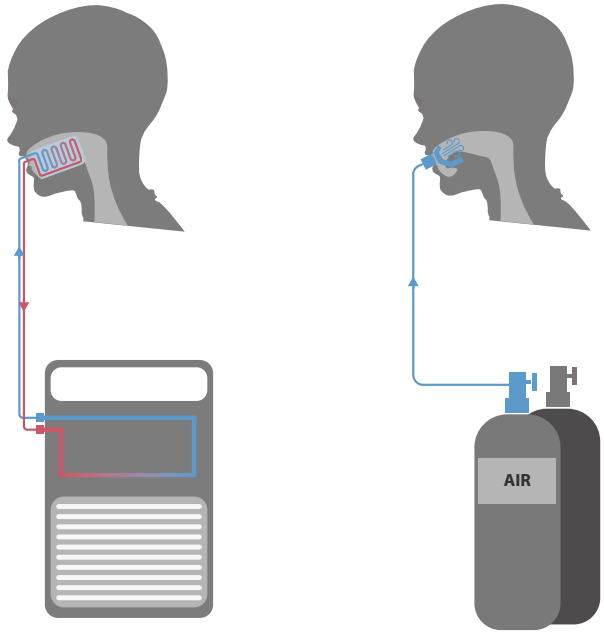
Using dry air would be the best way to prevent frost from forming inside the cooler, since moist air would condense and eventually form a clog in the system. To prevent the mouth from becoming too dry from the air, Hans and Peter suggested re-moisturizing the air with vapor or direct it through a small reservoir of water. The water could also be sprayed directly into the mouth. Another option would be to cool the air through expansion. The air would be kept in a liquid state with high pressure inside a tank and then expand when released. The phase change from liquid to gas would then cool the air.

Furthermore the low enthalpy of air would make the cooling process faster than with e.g. water, making the system more responsive and easier to manage.

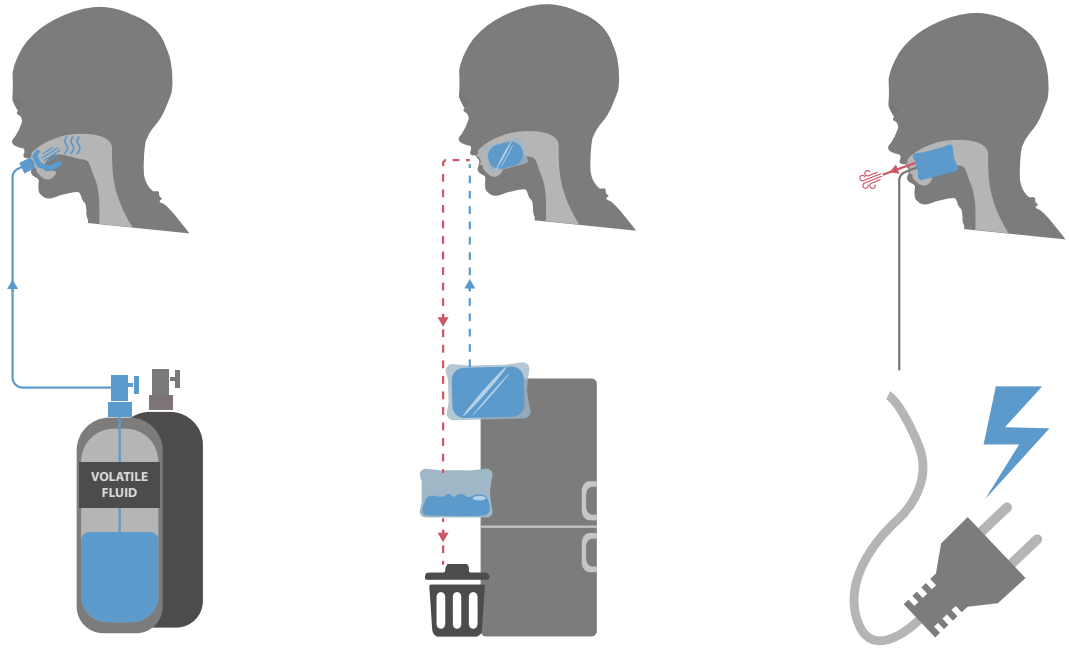
3.5.2 Principle evaluation

The evaluation is based on the findings presented in chapter 2.6 *Cooling techniques* and te result can be seen in figure 43. Evaporative cooling and thermoelectric cooling were found to be the concepts with the least technical feasibility and was therefore excluded from further evaluation. The thermoelectric cooling principle would generate a lot of heat which would have to be transferred outside the mouth resulting in a more complicated design.

The refill of volatile gas tubes would make the evaporative cooling principle both expensive and labor intensive, (National Institute for Health and Care Excellence, 2014).



	Closed loop water cooling	Open air cooling
+	Adjustable temperature Low labor (closed system) Common in med-tech	Adjustable temperature Non contact cooling Potentially small Universal mouth piece
—	Two thick hoses	May require refill Noise



Evaporative cooling	Passive cooling	Thermoelectric cooling
Non contact cooling Potentially small Universal mouth piece	Allows closed Mouth Cheap Small form factor Portable	Adjustable temperature Low labor(solid state) Silent
May require refill Noise	No temperature adjustment Labor intensive	Requires extensive heat transfer

Figure 43. *Illustrations of each cooling principle with accompanying evaluation.*

Cooling unit visualization

A quick ideation of how the remaining concepts could be embodied into a mouth cooling device was performed, see figure 44. The purpose of the visualisation was to bring attention to practical matters that needs to be taken into consideration for further evaluation. The ideation included factors such as labor intensity, practical integration and patient mobility

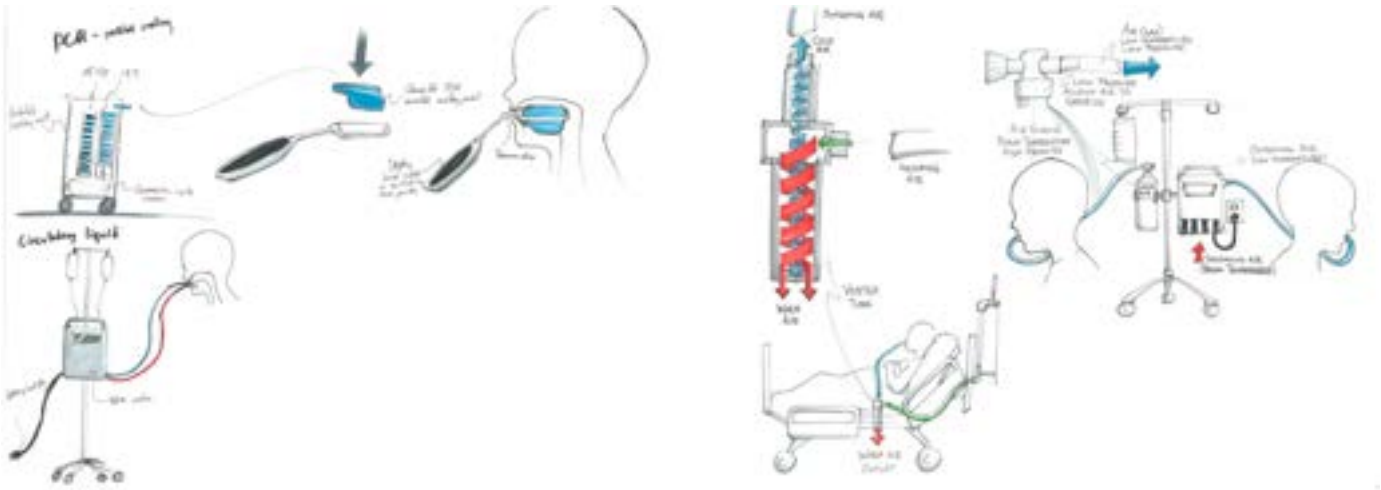


Figure 44. *Cooling unit sketches.*

3.5.3 Cooling principles tests

Quick tests were performed as a way of evaluating the cooling effect and comfort of the three remaining concepts, open air cooling, closed loop water cooling and passive cooling. The tests were self experienced by both authors of this report and did not include any external test subjects.

Passive cooling (PCM)

Cooling technique: Melt cooling
Material: 2 bags of frozen phase changing material with a melting point of 4°C.
Cooling time: 15 minutes
Surface temperature: 4°C

Execution: Two bags containing frozen phase changing material (PCM), were placed between the cheek and teeth on both sides of the mouth, see figure 45. When the content was melted the bags were replaced with new ones.

Result

+ Comfortable since the bags conforms to the shape of the mouth,
+ Cooling felt less intense than ice treatment
+ Do not require to keep mouth open

- Do not hold as much energy as water and has to be replaced every 5 minutes
- Uneven cooling
- Requires direct contact which makes it hard to cool e.g. soft palate

Mucosal temperature: 33°C



Figure 45. *Top, loading of PCM in small pouches. Bottom, Insertion of PCM pouch.*

Closed loop water

Cooling technique: Closed loop water cooling.
Material: 2 buckets, hose, mouthpiece, ice and water.
Cooling time: 15 minutes
Water temperature: 0°C

Execution: A hose was filled with water and ice, to generate a water temperature of 0°C. The hose was then connected to the bucket below the surface of the water. By diminishing the pressure in the hose a flow of water was created through the mouthpiece, see figure 46.

Result
+ Cooling felt less intense than ice treatment

- Cooling felt more intense than open air and melt cooling
- Uneven cooling
- Requires direct contact which makes it hard to cool e.g. soft palate

Mucosal temperature: 27,9 °C



Figure 46. *Water cooling test.*

Open air cooling

Cooling technique: Open air cooling.
Material: bucket, 2 hoses, bucket with lid, mouthpiece, pressurized air, ice and water.
Cooling time: 15 minutes
Air temperature: 4,6 °C

Execution: A bucket was filled with water and ice. Pressurized air was connected to the bucket below the surface forcing it to flow through the water. A hose with a mouthpiece was connected above the surface. The gap between lid and bucket was taped shut, forcing the incoming air to exit through the hose connected to the mouthpiece after being cooled by the cold water, see figure 47.

Result
+ Cooling felt less intense than ice treatment and closed loop water cooling
+ Perceived as the most comfortable cooling technique
+ Enabled cooling of more sensitive parts of the mouth

Mucosal temperature: 29,1°C

Conclusion
Since the bags of PCM used for melt cooling had to be replaced every 5 minutes this principle was deemed unfeasible and was ruled out from further evaluation. Air cooling was more comfortable than closed loop water cooling but the air did not reach the desired temperature and further tests had to be made.



Figure 47. *Air cooling test.*

3.5.4 Further validation open air cooling

Further tests of the air cooling principle were conducted to examine the feasibility of the principle.

Vortex tube

To further test the air cooling principle and in the same time test one of the concepts from previous cooling principle ideation in chapter 3.5 *Technical development* a vortex tube was created. The technology behind the vortex tube is described in chapter 2.5 Refrigeration technologies.

Cooling technique: Open air cooling.
Material: Vortex tube, 2 hoses, mouthpiece, pressurized air.
Cooling time: 30 minutes
Air temperature

Execution: A hose with pressurized air was connected to inlet on the top of the vortex tube, see figure 48. The other hose connected to the outlet on the cold side of the vortex tube, creating a flow of cold air through the mouthpiece connected to that hose, see figure 49.



Figure 49. *Vortex tube prototype prior to assembly.*

Result

The required airflow to acquire air in the right temperature was too high to create the right settings for the test. Furthermore the vortex tube created a really loud noise which led to the conclusion that the vortex tube concept should be excluded from further evaluation.



Figure 48. *Vortex tube prototype during testing. The tube has been attached to a vice as a safety measurement to avoid extreme temperatures.*

Further testing

Cooling technique: Open air cooling.
Material: 1,2 meter long closed drain pipe, bucket, 2 hoses, mouthpiece, ice and water.
Cooling time: 30 minutes

Execution: The test was set up in the same way as the previous open air cooling test, with the exception that the bucket was replaced with a 1,2 meter long drain pipe to extend the distance of cold water through which the air flowed before exiting the tube, thus creating colder air, see figure 50. The airflow was set to 40l/min, producing an air temperature of 3°C.

Result

The test confirmed the perception of open air cooling being more comfortable than closed loop cooling. The test resulted in a mouth temperature of approximately 23° C. The temperature was measured on the cheeks on both sides inside of the mouth.

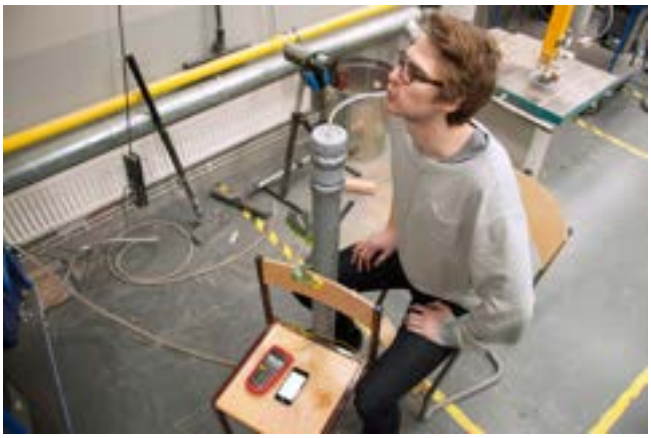


Figure 50. *Improved air cooling test.*

Stronger evidence

In a study investigating coronary responses to cold air inhalation during exertion, two experiments were conducted in which the participants breathed cold air between a temperature of -20° C and -24° C. The perceived thermal sensation and pain of the mouth were then rated. Where the thermal sensation was rated on a scale between 0 and 11, 0 meaning no sensation and 11 unbearable cold and the mouth pain on a scale between 0 and 10, 0 meaning no pain and 10 unbearable pain. The result of the experiments from the test groups from both experiments, which were not given any anesthesia, showed a rated mean thermal sensation of 2 and mean mouth pain of 0. The mean mouth temperature of the first experiment was measured to 13° C and in the second 17° C, (Muller et al., 2014). The result showed a low degree of pain and discomfort from the cold, suggesting that oral cryotherapy through air cooling is not perceived as particularly demanding even at air temperatures as low as -24° C.

Cooling unit - technical feasibility

To plan the final prototype and estimate the size of the cooling unit, the total power needed to produce air with the required temperature and airflow was calculated with equation 1 shown below.

$$Q = \Delta V \cdot \rho_{luft} \cdot c_p \cdot \Delta T \qquad (1)$$

Calculations were done for two cases. Case 1 with an airflow of 40 l/min and air temperature of 3° C, the same as in the last test of open air cooling, from subsection *Further testing*, which generated a mouth temperature of approximately 23° C. In case two, a margin of error was added to both parameters to obtain a maximum value for the prototype. The airflow was estimated to 80 l/min and the air temperature to -5° C. Explanation of parameters and their values in the two cases are shown in table 1 and the results from the calculations are shown in table 2.

3.5.4 Cooling principle decision

Passive cooling was evaluated as the least feasible principle because of the impracticality of constantly having to change the bags of PCM. Open air cooling and closed loop water cooling both proved to have enough cooling capacity. Since air cooling was more comfortable and better suited to cool the sensitive parts of the mouth it was decided as the principle best suited to proceed with.

Table 1. *The table shows an explanation of parameters and their values in two cases.*

Parameter	Description	SI unit	Case 1	Case 2
ΔV	*volumetric flow rate*	m^3/s	0.04/60	0.08/60
ρ_{luft}	*density air*	kg/m^3	1.3	1.3
c_p	*heat capacity air*	$J/(kg \cdot K)$	1004	1004
ΔT	*temperature difference*	K	17	25

Table 2. *The table show the results from the calculations.*

Result	Case 1	Case 2
Required power, Q	14.8 *W*	43.5 *W*

3.6 Ideation

After establishing a technically feasible foundation for the project the main ideation phase began, see figure 51. In the initial ideation ideas were brainstormed in a very unstructured fashion, drawing from knowledge and experiences from the literature and user studies. After the initial ideation, generated ideas were discussed and those with potential was brought further in the process to the targeted ideation phase. Three sets of ideation sessions were conducted on the topics of the design vision namely comfort, control and distraction.



Figure 51. *Ideation process*

3.6.1 Initial ideation

As shown in figure 52, the initial ideation resulted in ideas covering a broad spectrum. The ideas are illustrated using quick thumbnail sketches and the abstraction level is intentionally kept high to not get stuck in a specific direction. These ideas worked as a basis to build further on and concretize in the targeted ideation.



Figure 52. *Post-it notes from initial ideation*

3.6.2 Targeted ideation

Control

During ideation on the topic of control the concept of giving the patient a physical control unit, as seen in the initial ideation, was explored further. The ideas were structured based on output and input from the control or cooling device that is directed towards the patient, parent or both, see figure 53. An theme that influenced the session was to make the device and its actions understandable to the patient and parent.

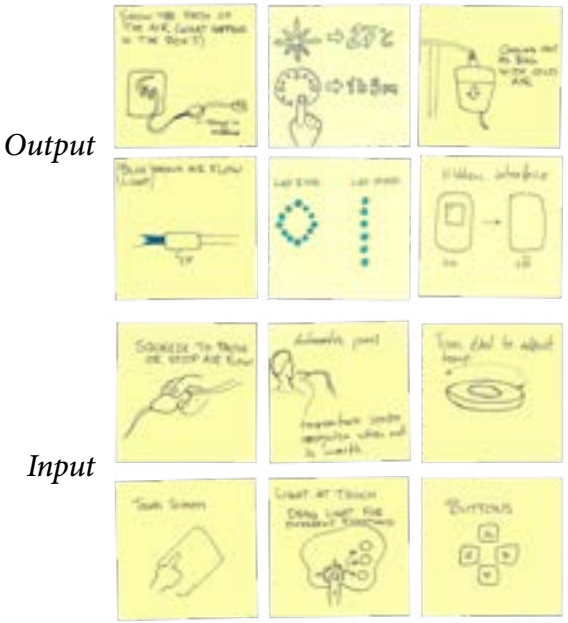


Figure 53. *Ideas from targeted ideation on creating control*

Distraction

The distraction ideation began with a lot of focus on how to turn the treatment into an entertaining game, see figure 54. There was however difficulties to create clear connections between the game and the progress and feedback from the treatment, something considered vital to make the game relevant. The focus was shifted to less intensive interactions such as light, haptics and graphical, inspired by an idea from the initial ideation of stimulating multiple senses.

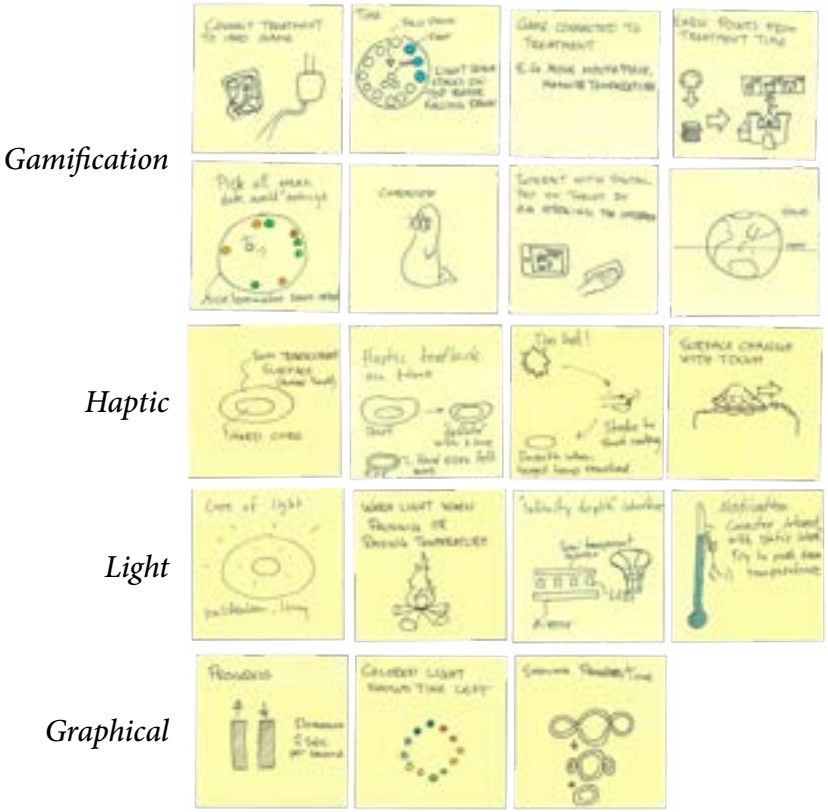


Figure 54. *Ideas from targeted ideation on distraction*

Comfort

The comfort ideation was mainly focused on hose management and the physical comfort, see figure 55. Having a mouthpiece with an air hose sticking out of the mouth will most likely limit the movement of the patient. Here the mission was to come up with ideas on how to minimize the discomfort of the hose and trying to make it as "invisible" for the patient as possible. Special attention was brought to allowing free movement of the head so that the hose do es not pull the mouthpiece when the patient turns their head, as well as providing a natural resting position for the mouthpiece when temporarily taken out.



Figure 55. *Ideas from targeted ideation on comfort*

3.6.3 Idea evaluation

Ideas from previous sessions were evaluated and decisions were made on which ideas to bring further, see figure 56.

Control

The need for patients to feel control over the treatment was first discovered in the literature study, see section 2.1.2 *children and cancer*, and then confirmed during the hospital visits, see chapter 3.1 *User understanding*, where a patient controlled morphine pump was presented. With this in mind the decision to base the patient experience around a patient control unit was taken. The choice of interface i.e. a touchscreen or simple light diodes with a dial, determines the direction of the patient control unit interaction. Hence, a mind-map over control interactions was constructed following the ideation phase which is presented in following section 3.6.4 *Patient interface framework*.

A potential issue with the patient control unit could be that parents might be worried that their child operate the device in the wrong way even if the patient control is restricted to only control the cooling unit within a set interval. To include parents in the treatment a decision was taken to go forward with an "ambient interface". The idea is that a simple light interface, clearly visible from the whole treatment room, should allow parents read the status of the treatment at a glance.

The hidden interface idea was taken from the initial ideation. It was mainly aimed for a light-diode based patient control unit where it may serve a decluttering function by hiding functions that are not in use. The hidden interface was also considered to contribute with both distraction and comfort. Distraction in that it adds to the "soulification" that is discussed further in the section below, and comfort given that it minimize the attention the product demands when not interacted with.

Distraction

It was decided to discontinue the gamification track. Although games can evoke a high level of engagement it was deemed unlikely that a game built in, or connected to, a mouth cooling device could compete with those already available on smart phones or tablets. From the newly formulated need of distracting without interfering with existing distractions came the idea to fascinate rather than entertain. As opposed to an entertaining game stimulating an active response the direction was turned against trying to replicate the soothing response of fascinating stimuli such watching as a fireplace or fish in an aquarium. That way the distraction is not actively calling for the child's attention but is still available if no other natural distractive element are used.

As a means of creating fascination, the ideas of "soulifying" and addressing multiple senses was brought from the initial ideation and elaborated into ideas of giving the control a silicon like material with a hard core that can be felt through the softer outside. The soft touch was also deemed important since the control most likely will lay in the bed with the patient. The idea of using a translucent material and light to create a glowing core was also brought further in the process with the ambition of creating the impression of something transforming and living rather than static.

Comfort

Gradual cooling and the ability for the patient to independently pause the air flow are two ideas taken directly from the initial ideation. These are considered essential to the product as they could solve the main comfort issues with the present ice treatment namely the entry point, see section 3.1.4 *User interviews*. The pause feature was considered important since it allows the patient to pause the airflow temporarily instead of aborting the treatment. Gradual cooling is a way lowering the threshold for how much cooling the patient must endure. It is based on the fact of the often overlooked difference between heat and temperature. It is the difference in temperature between the cooling medium and the oral mucosa that determines the heat transfer and thereby sensation of heat or cold. In other words, if your hands are ice cold you will perceive less cold water as warm despite its relatively low temperature. By gradual cooling the temperature difference between the oral mucosa and the air can be held low but constant to avoid an initial cold shock.

In terms of hose management three ideas were brought further. The three version of attaching the hose near the patient in order to prevent the weight from the hose to pull the mouthpiece was tested with quick mock-ups, see section 3.7 *Concept generation*. The idea of having a resting place for the mouthpiece was also brought further in the process.

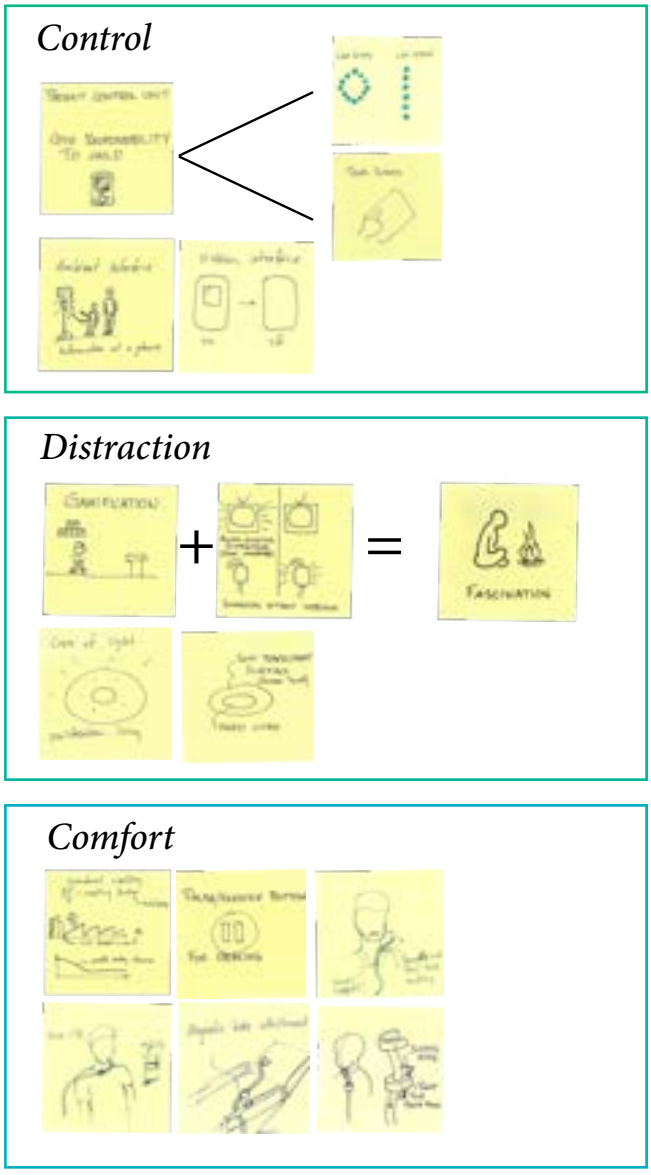


Figure 56. *Ideas brought forward from the ideation phase.*

3.6.4 Patient interface framework

To decide between a screen or light-based output on the patient control unit, a mind map over potential functions was made. Following a discussion the most necessary functions, highlighted in the mind map in figure 57, was chosen in order to not over-complicate the interface. Following the same rationale a LED-based interface was chosen due to its simplicity and potential to fascinate a young user group. During the discussion it was also questioned if the output and input have to be located on the patient control unit. This resulted in the idea of splitting the interface by placing the output on the cooling unit and thus minimize the size and weight of the separate patient control unit which would then only hold the input i.e. buttons or a dial.

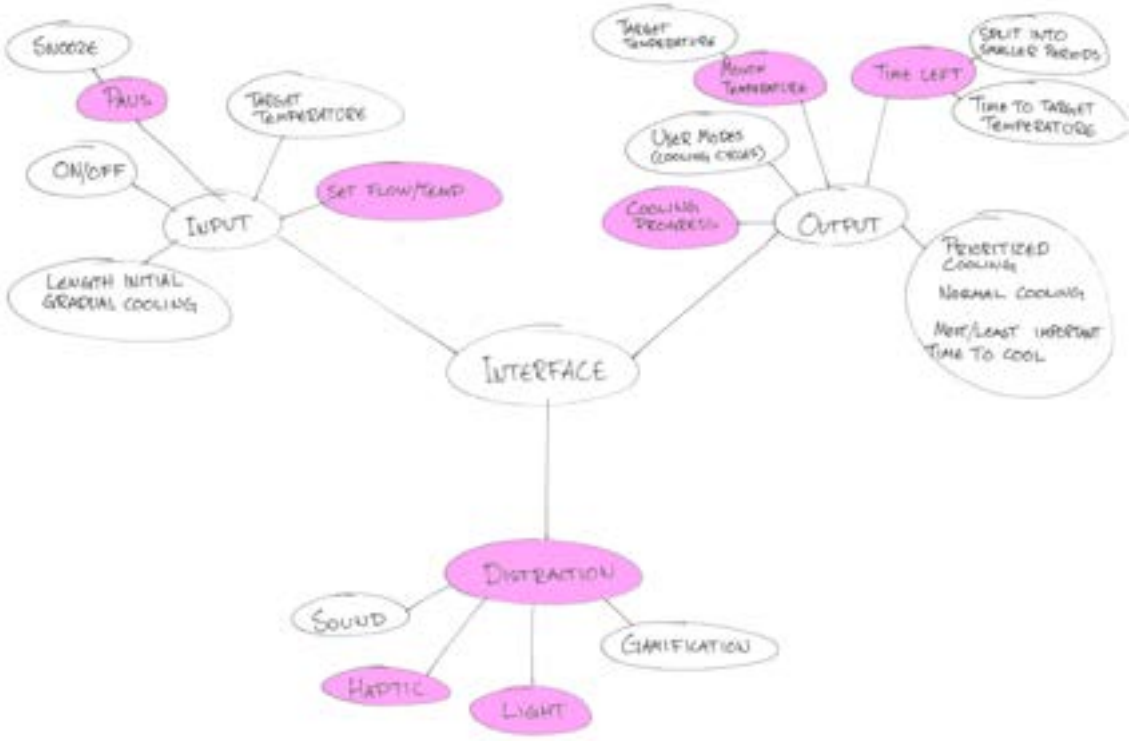
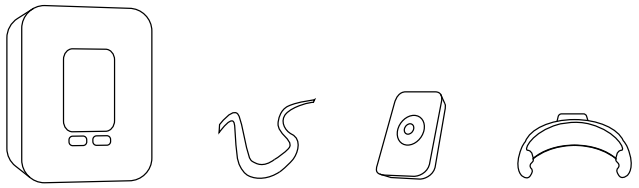


Figure 57. *Ideation on which information and functions the interface could provide. Selected ideas are highlighted in pink.*

3.6.5 Layout of components

In order to transition from a wide variety of ideas on separate features to generating concepts, there was a need to construct the final layout for the main components of the device. The decision to use open air cooling limits the final design to at least four fundamental components, see figure 58 below.



Cooling unit *Hose* *Patient control* *Mouthpiece*

Figure 58. *The four main components of the device.*

The position of the interface was thought to greatly influence the level of interaction with the device. With the ambition to provide a sense of control as well as distraction through the interface, a list of important factors presented below was made as a guiding reference for the layout.

- Interface should be readable with minimum effort
- Easy to reach and control device
- Minimal restriction of resting position
- Allow mobility
- Close enough to patient to provide distraction
- Positioned so that patient "owns" the control

Cooling unit position was also included in the ideation considering that the patient control interface and cooling unit does not necessarily have to be separated. A guiding list of important factors for placement of the cooling unit is presented below.

- Minimize noise from cooling unit heard by patient.
- Mobility
- Protect patient from cooling unit waste heat
- Accessible to nurse (in order to set treatment parameters)
- Easy mounting

Body worn

Although new technology within micro size air pumps and coolers could make the cooling unit reasonably light weight it would probably limit the performance, especially in terms of battery time. Primarily designed for an active use case, the body worn device would likely serve best when standing up or walking around. Based on interviews and observations most patients spend a majority of their time during chemotherapy in bed. Other issues with this alternative includes noise level, since the cooling unit is close to the user, as well as making sure the heat from the cooling device is directed away from the patient. Hence this alternative was ruled out.

Bed mounted

Including both bed mounted cooling unit and patient control unit interface these ideas rely on nurses mounting the device correctly on the bed, something they are not currently used to. The cooling units position was also deemed ergonomically unsound for the nurse who will have to bend down to access it. There are also problems with mobility since it requires the device to be either hand-held or remounted on to the iv pole.

IV-pole mounted

The three layouts based on an IV-pole mounted cooling device, see lower three sketches in figure 59, was chosen for further exploration. The cooling units placement on the IV-pole allows familiar mounting procedure for nurses as well as mobility for the patient. Another commonality is that all solutions allow the user have the pationet control unit within reach. This was considered especially important since it has to be clear that the patient control is meant for the child to interact with. It is also sought to create a sense of ownership of the control for the patient so that even though parents may partake in discovering the functions of the control, the child should be the only one who physically interact with the patient control unit.

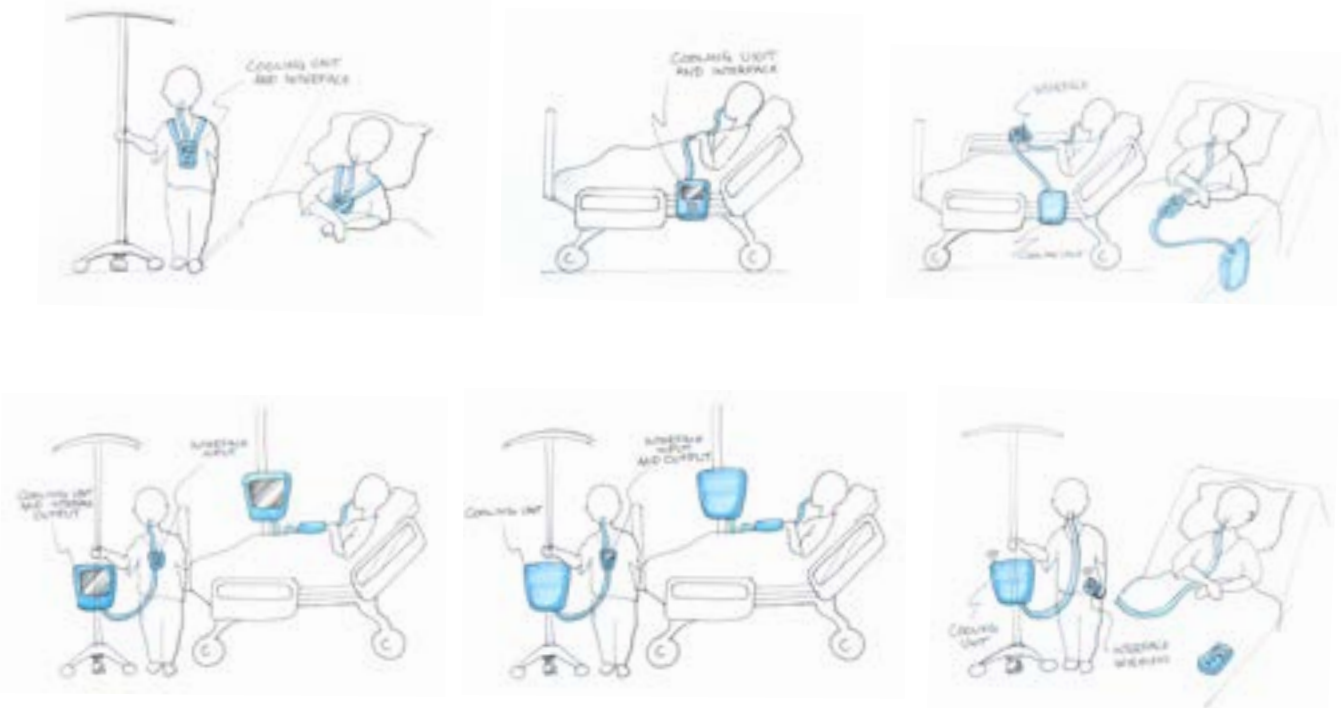


Figure 59. *Sketches on underlays with varied layout of components. The three bottom sketches represents the pole mounted concepts that was taken forward.*

3.7 Concept generation

In order to move forward after selecting three layout versions the design had to be further detailed to the point where mock-ups can be made and tested by the end user. This chapter entails the process from making the first explorative sketches of the patient interface , see figure 60, to four mock-ups that were tested and evaluated at the hospital, see chapter 3.8 *User testing*. As decided after the ideation phase, the patient control unit will be central in each concept and is thus the main focus at this stage. While the cooling principle and position has been decided, other remaining details that does not immediately affect the treatment experience for the patient are addressed in chapter 3.9 *Concept refinement*.

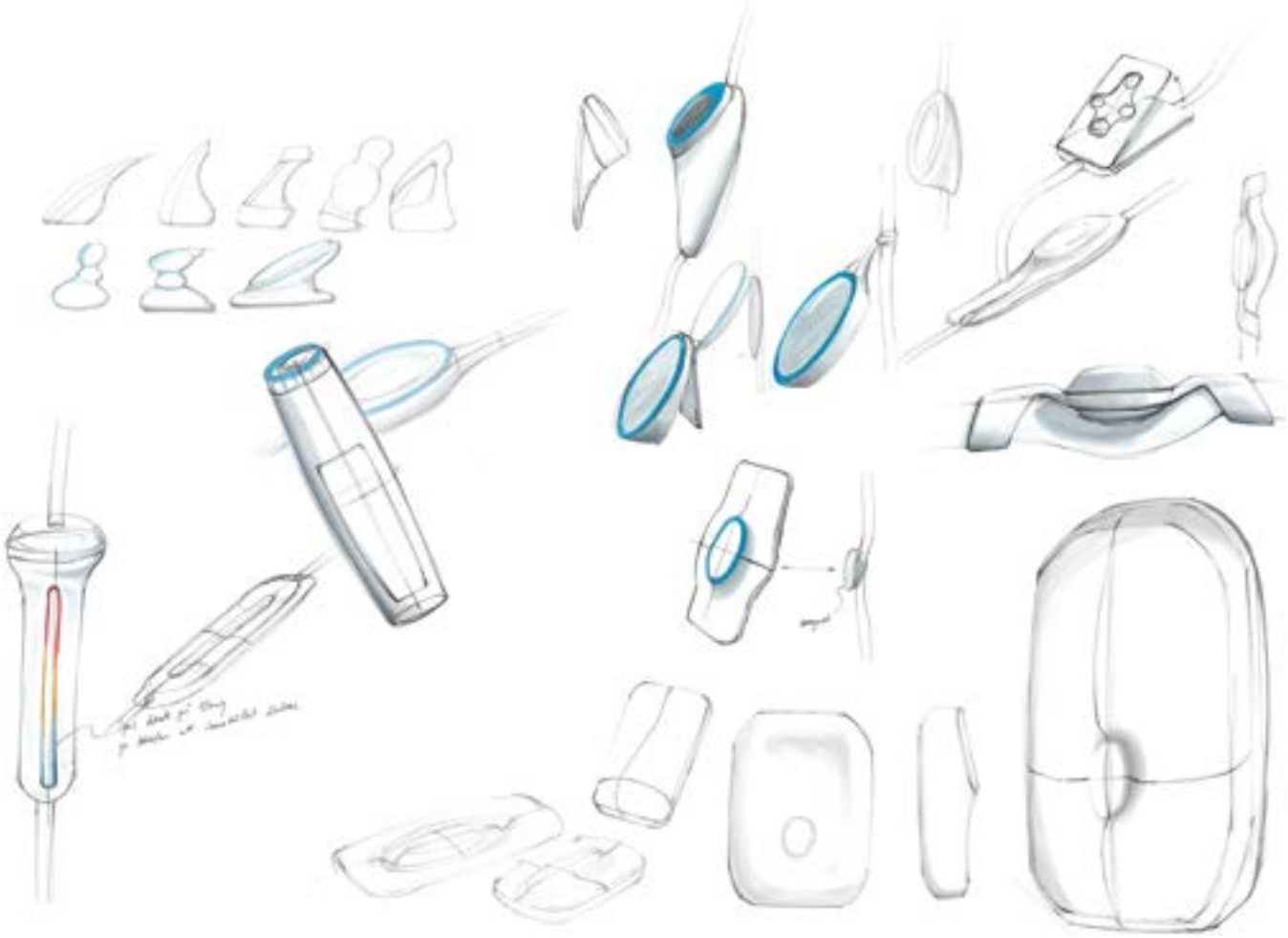


Figure 60. *Explorative sketches on the patient interface. Most sketches are of hand held patient control units and in lower right the sketches features interface output on the cooling unit.*

Explorative sketching

The main objective of this phase was to search for interface designs with good mapping i.e. an intuitive relationship between the interface input e.g. buttons, turning dial and their movements or effects. It was also considered how the patient control unit should be held and ways of turning the interface towards the user without bending the hose, thus minimizing effort.



Figure 61. *Mock-ups of patient control units and hose management ideas.*

Mock-ups

The sketches in previous section was evaluated and discussed after which mock-ups of the patient control unit was sculpted in foam material using simple hand tools, see figure 61. Mock-ups of ideas on hose attachments from the initial ideation was also created to get a rough estimation on how they could work in reality.

Concept direction

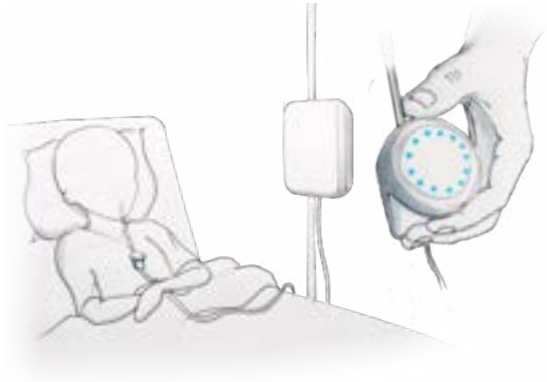
After testing and discussing the foam mock-ups of the patient control unit it was decided to go forward with an interface based on an LED-ring with a combined turning dial and push button. This design was chosen for its good mapping,- simplicity and small size.

The mapping of the chosen design was considered superior to the alternatives since the turning dial can be placed over the LED-ring. It gives the ability for the movement of the dial (input) to correspond directly to the light of the diodes (output) displayed on its surface. In other words, turning the dial one step clockwise would light up one diode in the same direction. This level of intuitive mapping could not be achieved with light strips and push buttons for instance. The dial was also considered to have an intuitive quality when it comes to regulate airflow seeing that it bares resemblance to other familiar actions such as adjusting water flow in a faucet or music volume on a stereo. The simplicity comes from the limited number of buttons due to the dial which can both be used to switch between functions and regulate air flow, something that would require two push buttons. This leads to the smallest form factor from any of the patient control alternatives that features both input and output on the control.

It was also decided to continue with the idea of a separated interface where the patient control unit only functions as an input device and the output is displayed on the front of the cooling unit. The choice was motivated by the small form factor allowed for the patient control unit which was thought to improve comfort.

3.7.1 Concepts

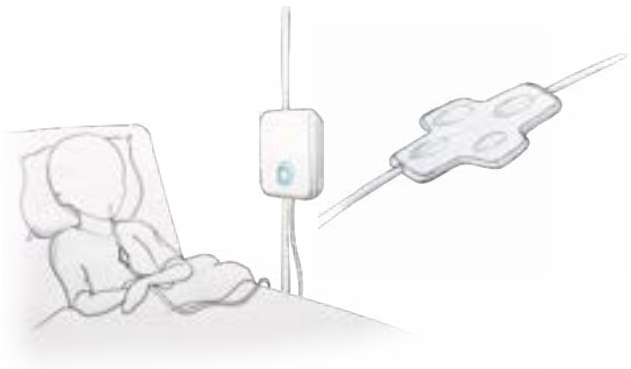
Four concepts were developed and later tested with mock-ups to asses which direction would fulfill the goals best. As described in previous section there are two interface directions, one with LED-ring and dial on the patient control unit and one with only button on the patient control unit and output on the cooling unit. Note that the latter lacks specifications on how the interface output on the cooling unit should be structured. To avoid wasting time on designing and building a separate mock-up of the interface output on the cooling, it was decided that a symbolic interface output for this concept was sufficient at this stage. The four concept are presented in figure 62 below. The rationale for including each of the concepts are presented in the pros and cons list describing expected benefits and drawbacks the concepts.



Concept 1: Tilted interface

\+ Does not bend hose
Peripheral visibility

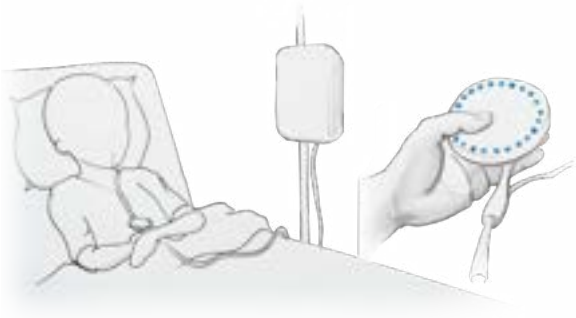
\- Must be attached to patients chest



Concept 2: Split

\+ Size
Weight (does not pull the hose)
Output also visible by parents

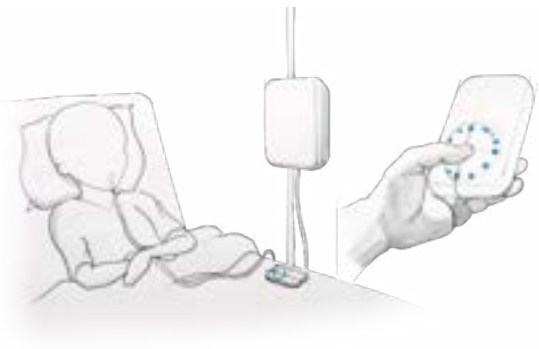
\- Less immersive interaction
Control mapping



Concept 3: Rubber strap

\+ Does not bend hose
Small

\- Hangs loose, risk getting in the way



Concept 4: Wireless

\+ Easy to put away
Does not affect hose management

\- Risk of loosing control
Reduced feeling of ownership

Figure 62. *Sketches of four concepts with accompanying list of expected benefits and drawbacks.*

3.8 User testing

During the ideation process it became evident that the placement of the interface would be a determining factor for its design. Since the main purpose of the interface was to function as a distracting element and to give a sense of control, it was considered important to make it easily accessible for the patient. It was hypothesized that placing the interface close to the patient would give a sense of ownership and maximize accessibility to the interface input and output. However, placing the interface too close could restrict or disturb the patient rather than creating a more pleasant treatment experience.

Four mock-ups, representing each of the concepts in previous section, was built using a common interface module. The mock-ups were tested and evaluated by the participants, while being observed and interviewed. Particular attention was brought to observe the overall response to a light-based display.

Interface mock-up

Four patient control units in scale 1:1 was made of polyurethane foam and used as mock-ups to test different locations and shapes, see figure 63. The interface was made from an adafruit neopixel 16x LED ring with a push button mounted in the center of the ring. Due to time limitations the rotational function of the button was omitted in the mock-up. The LEDs were covered by a rubber like surface made from Sugru moldable glue. The LED ring was run by an Arduino uno microprocessor programmed in C++. The size of the cooling unit was roughly estimated and a 1:1 scale model in polyurethane foam was made to be attached to the iv-pole.

Preparatory user test

Before performing tests with patients at the hospital, a preparatory pre-test was conducted to evaluate the mock-ups and test procedure, see figure 64. It was deemed especially important that the hospital test could be carried out in a well structured manner since it would be performed with sick children and limited time. The pre-test was performed at the Royal Institute of Technology in Stockholm and included two adult participants. Both participants tested all of the four concept mock-ups while being observed and interviewed in an unstructured manner. Although the test functioned as a preparatory pre-test, the results from this test were also taken into account when evaluating the concepts.



Figure 64. *Photo taken during preparations for the pretest.*



Figure 63. *A collage of the mock-up process and the resulting models that was used in consecutive tests.*

3.8.1 User test at the Hospital

Following the preparatory pre-test a new user test was conducted at Uppsala University Hospital. One patient participated in the test, while a nurse and a grandparent to the patient acted as support.

The test was conducted in the patient's room at the hospital. A transparent plastic hose was used to simulate the air delivery hose. The hose was drawn from the cooling unit to the chest near the mouth of the patient. No air was blown through the hose and the mouthpiece was placed close to the mouth of the participant and not in it.

Stimuli

Due to both practical and regulatory reasons the interface could not be tested in a real scenario, where its function would allow the user to initiate and adjust the airflow as well as receiving feedback of the cooling progress. Instead a simple task was designed to have the participant interact with the interface in a purposeful way.

The participant was instructed to produce a specified number of light dots in a particular colour, as seen in figure 65. For instance, to produce five yellow dots, the child had to push

the button once to start a sequence fading between different colors of light, wait for the colour to turn yellow, push the button again and wait for five yellow dots to be lit before finally pushing the button a third time to stop the sequence at five yellow dots.



Figure 65. *Sheet of paper with different tasks for the test participant to complete.*

Procedure

The test started by setting up the equipment. The general design of the cooling device and its components were described using a child friendly illustration, see appendix B, in order to provide the participant with a context for the test. The participating patient was then presented with the interface and encouraged to play around with it for a while. Once the participant was accustomed to the device the test began.

The task, described in the section above, was repeated a number of times with the interface attached to one of the four patient control mock-ups. The child, parents and nurse was asked questions about the concept, after which the interface was attached to one of the other mock-ups. The process was carried out for each of the four concepts. After all concepts had been tested more general questions, were discussed with the child, parents and nurse. Observation guidelines and questions are shown in appendix A.

Results

This section includes results from both user tests, the pre-test and the test at the hospital. The results are categorized by concept.

Tilted control

The patient from the test at the hospital had no clothes on the upper body which excluded the option of attachment with the clip and allowed free placement of the control. The shape of the tilted control unit seemed to offer a good grip. Letting it rest on the stomach while holding it in place with one hand and using the other to press the button. This concept seemed to be the easiest to handle for the patient, who also nominated it as the favorite concept.

Participants from the pre-test appreciated the control when sitting up but thought it was hard to see when lying down, requiring tilting the head more than what was comfortable. Both participants expressed that they would prefer if the control could be slided on the hose.



Figure 66. *Tilted control mock-up.*

Wireless control

Both participants from the pre-test thought it would be a good idea to be able to attach the wireless control to something while laying in bed, see figure 67. They also said that they liked the idea of being able to put it aside when not in use. When using the wireless control, which was the only concept not attached on a fixed level of the body, the participants from the pre-test held it with straight arms in level of the hip. Both of them ranked the wireless control as their favorite.

As well as the participants from the pre-test, the patient seemed to prefer holding the control unit with straight arms in level of the hip or lower part of the stomach. The patient ranked the wireless control unit as the second best concept, together with the rubber strap control unit.



Figure 67. *Wireless control mock-up.*

Rubber strap control

The participants from the pre-test liked the idea of being able to tilt the control unit without having to lift the hose, see figure 68. Because of the small hands of the patient, the size of the rubber band mock-up seemed to make it easier to hold than the larger wireless control. Despite the small mock-up, the patient held it with two hands. The patient ranked the rubber strap control unit as the second best concept, together with the wireless control unit.



Figure 68. *Rubber strap control mock-up.*

Cooling unit interface
The participants from the pre-test thought it would be helpful if the height of the cooling unit could be adjusted on the IV-pole, see figure 69. They also felt that it was comfortable with a smaller control unit while laying in bed. One of the participants thought it was easier to see the interface when placed further away.

It was difficult to test the input control since it did not have any working buttons which also seemed to make the patient less interested. The patient ranked the concept as the least favorite.



Figure 69. *Cooling unit interface mock-up.*

Evaluation with nurses

After the user tests the concepts were discussed with two nurses from the department, see figure 70. One of the nurses mentioned that patients often find it annoying to keep things in bed and believed that it would be a good idea to enable the possibility of placing the control outside of bed. She also thought it would be a good idea if the patient could move around with the device during treatment.

The importance of preparing the patient mentally for the treatment was discussed. The child needs to be familiar with the device and the parent will probably ask a lot of questions about the treatment, to know which type of supervision is required from them and if they will need to help the child with anything. Furthermore the diversity of users were discussed, that the device should be designed not to take more attention than needed for those who do not want it. Some might find flashing lights stressful or annoying. It is therefore important that they can turn them off, especially during night when they are trying to sleep.

Concerning the idea of a wireless control, it might seem nice from a patient's point of view, however it has to be made sure that the controls are not lost. The fact that the control can be left behind, switched or that some children might bring it home with them if they like it, has to be considered.



Figure 70. *Concept evaluation with one of the nurses.*

Discussion
It would have been preferable to have more participants to increase the diversity of the test result. Only one patient was able to participate, which might have affected the results.

The test showed the importance of a flexible solution, the patient spend a lot of time in bed and will have a lot of time adjusting the device to their liking if the design allows it. There should definitely be a simple way for the patient to put the control aside. Many patients choose not to have clothes on while lying in bed, which makes the clip meant to attach the control to the user's clothes redundant in those cases.

Both nurses and grandparent were positive to the idea of the child controlling the device. The nurses explained that they always strive to involve the children as much as possible claiming that children noticeably appreciate when they can partake and do things themselves.

"The child has no problems doing things themselves if you just explain and let them try it before" - Grandmother.

The patient seemed interested in the control, became absorbed by the task and showed a rapid learning curve. During the test there were some technical problems with the control, which occasionally did not register a press of the button. When this happened the patient tried to solve the problem rather than asking for help. The patient was noticeably distracted by the control and the task. This was indicated by the fact that all focus were kept on the control unit unless being addressed directly.

3.8.2 Concept direction

As presented in section 3.6.5 *Layout of components*, an IV-pole attached cooling unit was chosen. The test presented in previous section showed that the hand held light interface was appreciated by the child. The concept where the patient interface has the output on the cooling unit was therefore rejected.

A decision was made to attach the patient control unit on the hose. This is motivated by the fact that it is more practical for the nurse who will administer the treatment if every part of the device is physically connected so that there is no risk of misplacing anything. By attaching the control to the hose it also ensured that the control is always within reach for the patient and prevents parents from using the control in the place of the patient. The test clearly showed that its important that the control can be put aside. Thus, it was decided that the control should be able to slide on the hose. The tilted interface concept proved to be the most appreciated in the hospital test and is also in line with the other decisions. The shape was however was specifically designed to sit on the chest of the patient and be visible in the periphery. The shape will therefore have to be redesigned to fit a scenario were it is held closer to the hip. The illustration in figure 71 outlines the concept direction. The process of refining the design is described in the following chapter.

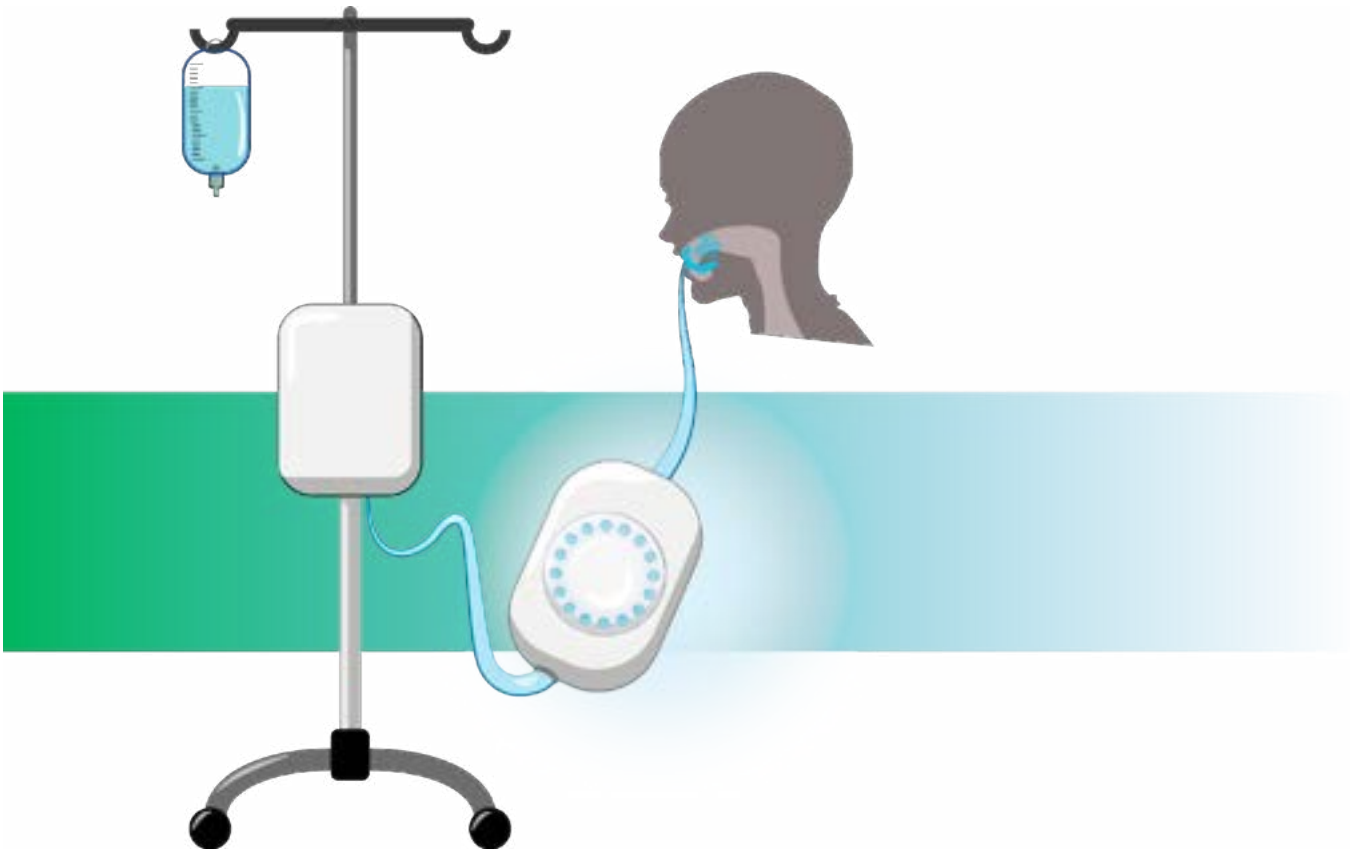


Figure 71. *Illustration of the chosen concept direction.*

3.9 Concept refinement

This chapter entails the concept refinement phase which cover the process from final concept to final design. The refinement of both patient control unit and cooling unit was carried out in parallel but are presented here in separate sections for the sake of readability.

3.9.1 Moodboard

In the beginning of the refinement phase a moodboard was constructed, see figure 72. The college is structured in two parts where the upper part communicates material and basic shapes and the lower part entails inspirational products.



Figure 72. *Moodboard*

3.9.2 Patient control unit

The patient control unit refinement processes began with finalizing the size and form of the control body and dial. The detailed design of how the unit connects to the rest of the product i.e. hose and power supply, as well as the patient interface package, was carried out concurrently but are presented in the following subsection. After shape and hardware design was determined the framework for the interface was constructed.

Size and form evaluation

The process of refining the form and size of the patient control unit began with line drawings i Adobe Illustrator see figure 73. The outlines were then transfered to CAD models using Rhinoceros 3D software to construct multiple models with slight variations in rounds and curves, as well as size. A number of shapes were selected and turned into physical foam models using a 3D mill. The foam models where evaluated and altered slightly using sandpaper. After reaching a satisfactory shape the CAD model was updated to correspond to the physical foam model and the shape was finalized.



Figure 73. *Refinement process of the patient control unit.*

Detailed design

In parallel to the form and size work, detailed design on hose attachment, power supply and connection to the cooling unit was performed.

Control attachment

As described in section 3.8.2 *Concept decision,* the patient control unit should be attached to the hose while still allowing the patient to adjust its position on the hose. In order to provide comfortable grip around the backside of the control it was decided that the hose should run through the control as opposed to on the outside. A certain amount of friction between the hose and control was needed to make the control stick at desired position on the hose. This was solved by letting the hose run in a slight turn around the center of the control which also leaves more room for interface electronics, see figure 74.



Figure 70. *Top view of the patient control unit, interface electronics and dial has been remover to show the hose arrangement.*

The decision to make the controls position on the hose adjustable meant power supply and data transfer could not go by cables. Thus the control will be powered by a rechargeable battery that is charged with induction trough the bottom of the control when docked on the cooling unit, see following section 3.9.3.*Cooling unit*. The data transfer will be handled by a small low-energy blue-tooth module.

Hose management

Given that patients does not always wear clothes on the upper body, another solution than attaching the control to the patients shirt had to be found. The resulting design shown in figure 75, utilize the same magnet on the back of the control used for docking it onto the cooling unit to attach it to the bed rail or other steel parts of the bed. The purpose of the attachment is to provide strain relief for the hose to prevent it form pulling the mouthpiece and also to make it easier to keep track of the control.



Figure 75. *Patient control unit attached to hospital bed rail by use of a magnet.*

Patient control unit interface

The detailed design of the patient control unit interface is shown in exploded view in figure 76. At the bottom of the build lies the push button encoder which registers both rotational input and downward pushes. Above the encoder push button is the plastic housing which also holds an LED ring and four independent LEDs in the middle. The LED ring act as the main output within each function menu of the control while the Four LEDs in the middle are covered by an overlay which projects the symbols of the four functions. The dial consists of a stiff transparent plastic shell covered with a layer of soft translucent silicon rubber.



Figure 76. *Exploded view of the patient control unit interface. From bottom, push button encoder, housing, LED ring, menu LEDs, menu overlay, transparent dial button, dial silicon cover.*

Control interface framework

The patient control unit features four functions or menus, pause, airflow, mouth temperature and treatment time, arranged in the center circle of the dial. For each function the information is given by the outer LED-ring, see figure 77. The interface is entered by pushing down the dial which wake the control up and displays the first function, pause. The dial can be rotated in either direction to skip between the functions. The circular arrangement of the function icons allows for good mapping between the rotating direction and the interface feedback.

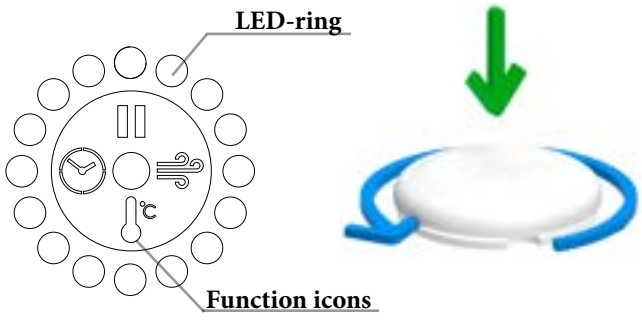


Figure 77. *Left, Schematic top view of the interface output. Right, interface dial and its two input actions.*

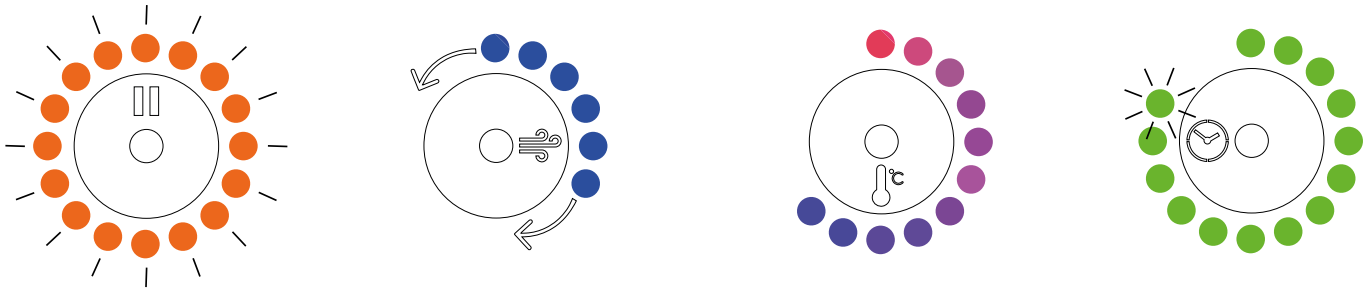


Figure 78. *The four functions of the patient control unit. From left, pause, airflow, temperature and time.*

Pause
The pause function stops the airflow temporarily and is easily accessed by pushing down the dial when the pause icon is displayed. The airflow will gradually start again after 30 seconds if the mouthpiece is still inside the mouth as indicated by the mouthpiece temperature sensor. While the pause is in effect the LED-ring pulsates in a warm orange color to signal that the mouth is getting warmer.

Airflow
Apart from the pause function, airflow is the only function in which the patient can adjust the cooling. When entering the function by pushing down the dial, blue LEDs show current level of airflow. By rotating the dial the user can increase or decrease the airflow. If the airflow is decreased to the point that target temperature will not be reached in time, the flow will only be decreased temporarily after which the device will gradually increase the flow to an appropriate rate. The span within which the airflow can be adjusted by the patient is also restricted. The purpose of the restrictions is to make sure diversions from the recommended treatment is intentional, therefore this can only be achieved by taking the mouthpiece out of the mouth.

Temperature
The temperature function shows the mouth temperature in relation to target temperature. The number of lit LEDs as well as the hue of the colour shows how far from target temperature the mouth temperature is. Once the target temperature is reached the circle is closed and pulsates slowly.

Time
Pushing down the dial when the clock icon is shown takes the user to the time function. The number of lights lit in the LED-ring indicate how elapsed time of treatment in relation to total treatment time. The number of light dots correspond to how much time is left, thus the time is counted down meaning the indicator starts as a full circle with the light currently about to be turned of pulsating, see figure 78.

Color, Material, Finish

The material of the control plays a central part in the final design. It has the function of spreading the light from the interface throughout the shape as well as accentuating the hard core of the control. A white translucent silicone rubber was chosen to provide these functions, see figure 79. The shape was designed with relatively small rounding with the intention to bring forward the soft properties of the silicon. The translucency of the material also contributes to edges being perceived rounder than they actually are because more light is let through where the material is thin.

A rough textured surface was chosen to make the control more haptically interesting and provide a soft touch which is important considering that the control will be in the bed with the patient. Aesthetically the control sought to be rather anonymous when not in use. It is when interacted with that the product comes to life through the colorful lights that spreads throughout the silicon material.



Figure 79. *Rendered image of the patient control unit in white translucent silicon rubber.*

3.9.3 Cooling unit

This section describes further development and refinement of the cooling unit. The refinement process starts with an overall form evaluation, followed by the detail design of specific components. The process then continues with the design of the interface and finally ends with a subsection concerning color, material and finish.

Form evaluation

The form evaluation process began with the generation of rapid sketches, seen in figure 80. The sketching process functioned as a way of discovering and trying out different shapes and implementations of functions on the cooling unit. Emphasis was on the quantity of sketches rather than quality, to create a comprehensive foundation of shapes to evaluate and choose from.

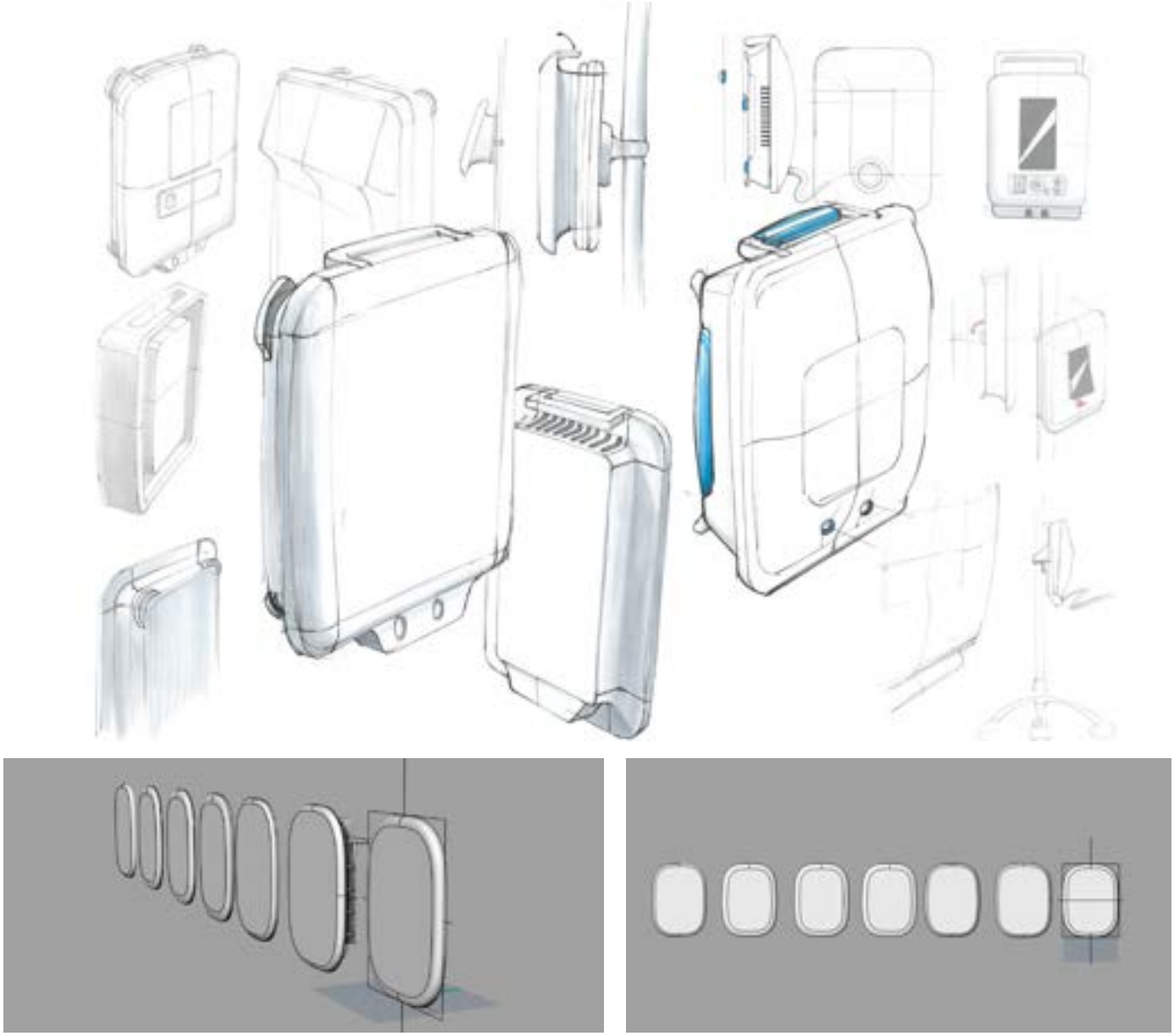


Figure 80. *Refinement process of the cooling unit's shape and general functions.*

A number of 3D-models were then created based on these sketches. The purpose of the models was to produce more detailed shape variations in a more apprehensible format, easier to grasp in the continuing evaluation.

From evaluation and discussion of the CAD-models the cooling unit's main shape was decided, see figure 81. The shape was designed to direct all the focus to the front of the cooling unit, where all the interaction with the device takes place. Round corners and flowing continuous surfaces are used to create a soft and friendly form language blending with the surroundings, following the overall design language.



Figure 81. *Main shape of the final cooling unit.*

Detail design

Ventilation holes were placed on the side to vent out waste heat from the thermoelectric coolers. Variations of shape and placement were generated in the 3D-model from the previous subsection and then evaluated, see figure 82.



Figure 82. *Variations of waste heat ventilation holes.*

All interactive elements were placed on the front panel to make the user interaction as easy and intuitive as possible. Nurse display, charging dock and air hose connections were laid out on the large flat front as tools on a workshop wall, to communicate simplicity and structure, see figure 83.

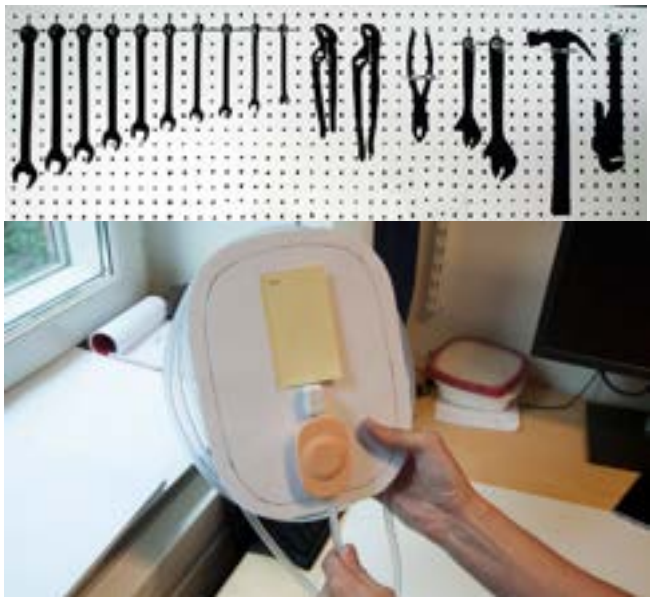


Figure 79. *Tool silhouettes on a workshop wall functioned as an analogy to the layout of components on the cooling unit front surface.*

A hose connection on the bottom of the cooling unit was designed, see figure 84. The hose connection was designed as a separate module protruding from the bottom of the casing. The idea of the design was to keep the front surface as clean as possible avoiding breaches and too much distracting details.

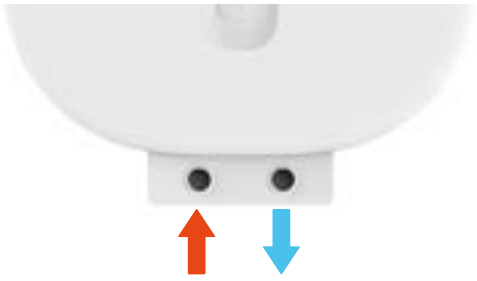


Figure 84. *Hose connector from front view, located at the bottom of the unit and separated from the flat front surface.*

An indent in the front surface was designed as a docking station where the patient control unit is attached with a magnet, see figure 85. The docking station is used for charging the control unit through induction and ease transportation of the entire cooling unit by connecting all parts.



Figure 81. *Left, control about to be docked onto cooling unit. Right, close up of docking indent showing battery charging symbol.*

It was decided that thermoelectric peltier coolers would be used as the refrigeration technology in the cooling unit.
The low weight and small size of the coolers motivated the decision, despite their low energy efficiency.

The coolers are connected to a metal tube with high conductivity which cools the pressurized air as it is blown through the loop. Excessive heat from the coolers are led to heat sink s which are cooled by small fans that pushes the hot air out through the vent holes on the side of the unit. To allow mobile use, external battery and air tank can be connected to the cooling and attached to the IV-pole. However this feature was moved to further development, see chapter 7 *Recommended future work*. Figure 86 below shows the internal layout of cooling components.



Figure 86. *The figure shows a suggestion of the cooling component layout where the black fans are connected to the grey heat sinks, peltier elements are shown in green and air cooling tube is the white part in the middle.*

Parent interface
An ambient light interface on top of the cooling unit was designed as a way for the parents to keep track of the treatment. The interface works as a reassurance for the parents as well as preventing them from using the patient control unit, making it more personal for the child. While turned off the interface is hidden, blending with the front surface of the cooling unit.

The color of the light indicates the status of the treatment, see figure 87. Every fifteen minutes a slow light pulse provides updates of the progress and signals if anything is wrong. Progress updates can be disabled if they are perceived as annoying. Constant orange light indicates that the treatment is paused and disappears when un-paused, red light indicates that something is wrong. Progress updates are indicated by a single slow pulse of light where blue means that the mouth temperature is acceptable and orange that it is too warm. The information allows parent to partake in the treatment by complementing or motivating their child.

Nurse Interface
An interface were designed for operating the cooling unit and to show more detailed information about the treatment, mainly meant to be used by the nurse. When turned off the interface is hidden, blending with the front surface of the cooling unit.

The interface was designed to be as simple to operate as possible, with an intuitive touch screen and a minimized amount of steps needed for navigation. Figure 88 show the nurse display after the unit has been connected to an a power source and figure 89 shows a wire frame of how the interface is navigated.



Figure 87. *Ambient interface mainly directed towards parents.*



Figure 88. *Start screen of the display shown immediately after the device has been connected to a power source. By pushing the user is directed to the menus on the home screen shown below.*

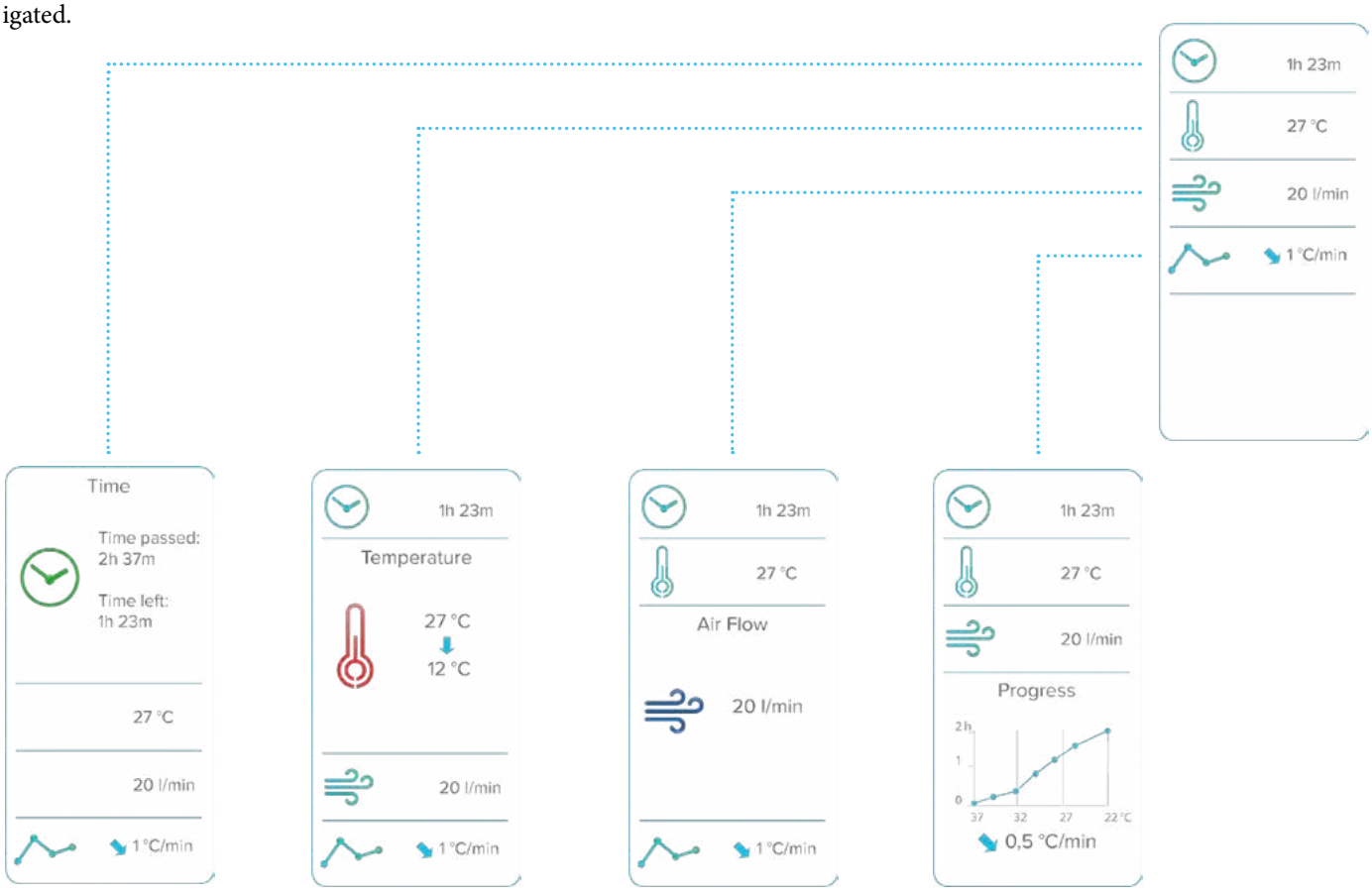


Figure 89. *Wire frame of the nurse interface and its four menus, time, temperature, airflow and progress.*

Color material and finish

The decision of color for the casing of the cooling unit was based on previous design guidelines, emphasizing on a clean and discrete design adapted to the context of a typical hospital environment. A matte white color was deemed as the best fit based on those guidelines and to match the design of the patient control unit.

Color variations of details such as the handle and hose connection module were evaluated, see figure 90. It was decided those details should have the same matte white color as the casing to keep the overall finish from demanding to much attention from the user and to direct the attention towards the patient control unit. Therefore the child friendly playfulness of the design was limited to the colorful light of the patient control unit.

A translucent plastic material were decided for the casing to allow the light interfaces to shine through. The final color and finish are shown in figure 91.



Figure 90. *Colour variations*



Figure 91. *Rendering of cooling unit with selected color and material.*

3.9.4 Mouthpiece

A mouthpiece that will deliver the cold air to the patient was designed in order to complete the product. Although this is a part that might have a large effect on the comfort of the treatment, it was deemed outside the scope of this project to conduct the comprehensive deign process required to design an optimized mouthpiece.

The design shown in figure 92 features three six air inlet holes on each side of the mouth and two under the tongue. The exhaust air channel in the front is depicted in red.

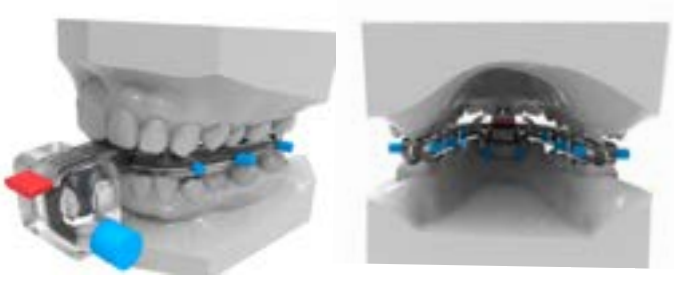


Figure 92. *Mouthpiece shown in transparent plastic inserted in mouth. Cold air inlet channels are shown as blue solids while hot air exhaust channel is shown in red.*

3.10 Prototyping

Prototypes of the cooling unit and control unit were made to facilitate user testing of the final concept. A functional cooling unit was also made to create cool air.

Cooling Unit

The process of making the prototype of the cooling unit began by milling two foam cores with a CNC mill, one for the backside and one for the front. Vacuum forming where then used to cover the cores with white plastic sheets, creating a plastic casing for the model. The foam cores where then glued together and cavities for LEDs were cut with the CNC mill to create room for the two light interfaces on the front and top of the cooling unit. Ventilation holes were drilled according to the designed pattern and the plate for the charging dock was cut with a laser cutter. Some of the smaller details and parts such as the handle and hose connector were made with a 3D printer. The collage in figure 93 below shows parts of the process.



Figure 93. *The figure shows a collage of pictures taken during the process of making the prototype of the cooling unit.*

Functional air cooler

The process of making the functional air cooler began by bending and drilling sheets of metal to create a frame for the components. A compressor with attached controller, air cooled condenser and a pipe heat exchanger to cool the air where then attached to the frame. Copper tubes and a capillary tube were used to connect the components by soldering them together and a refrigerant could then be added to the closed system. The compressor controller was programmed with external resistors to set the right rotation speed of the compressor before connecting the prototype to a battery. Parts of the process is shown in the collage in figure 94.



Figure 94. *The figure shows a collage of pictures taken during the process of making the functional air cooler prototype.*

Control Unit

The process of making a prototype of the control unit began by making cast molds for the control's base and the dial, where the shape and size was based on previous CAD models and milled with a CNC mill. To acquire the desired surface structure a foam material was used for the cast molds, transferring the porus surface of the foam to the silicone. Both control body and dial were then casted in white silicon rubber. The casing was made with a 3D printer adjusted to fit a rotary encoder with a push button, an Adafruit Neopixel 16x LED ring and four separate LEDs, which were soldered to a cable leading to an Arduino Uno microprocessor. The microprocessor controlling the interfaces were programmed in C++, see appendix C. A plastic overlay were then cut with a laser cutter to create symbols needed for the interface menu. The soft silicone button was attached to a hard translucent plastic plate before assembling the whole prototype. Parts of the process are shown as a collage in figure 95 below.

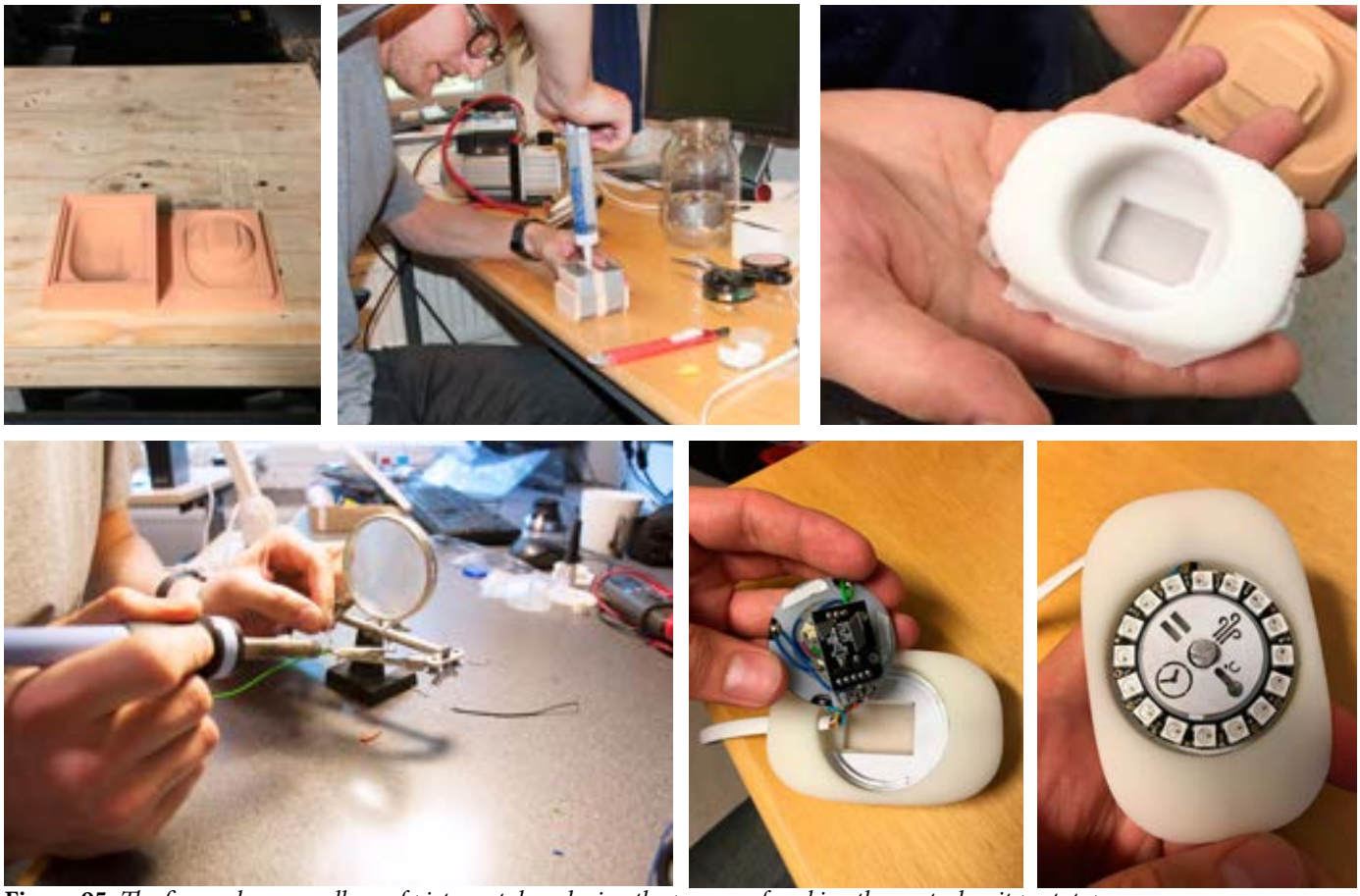


Figure 95. *The figure shows a collage of pictures taken during the process of making the control unit prototype.*

4. FINAL DESIGN





4.1 Function

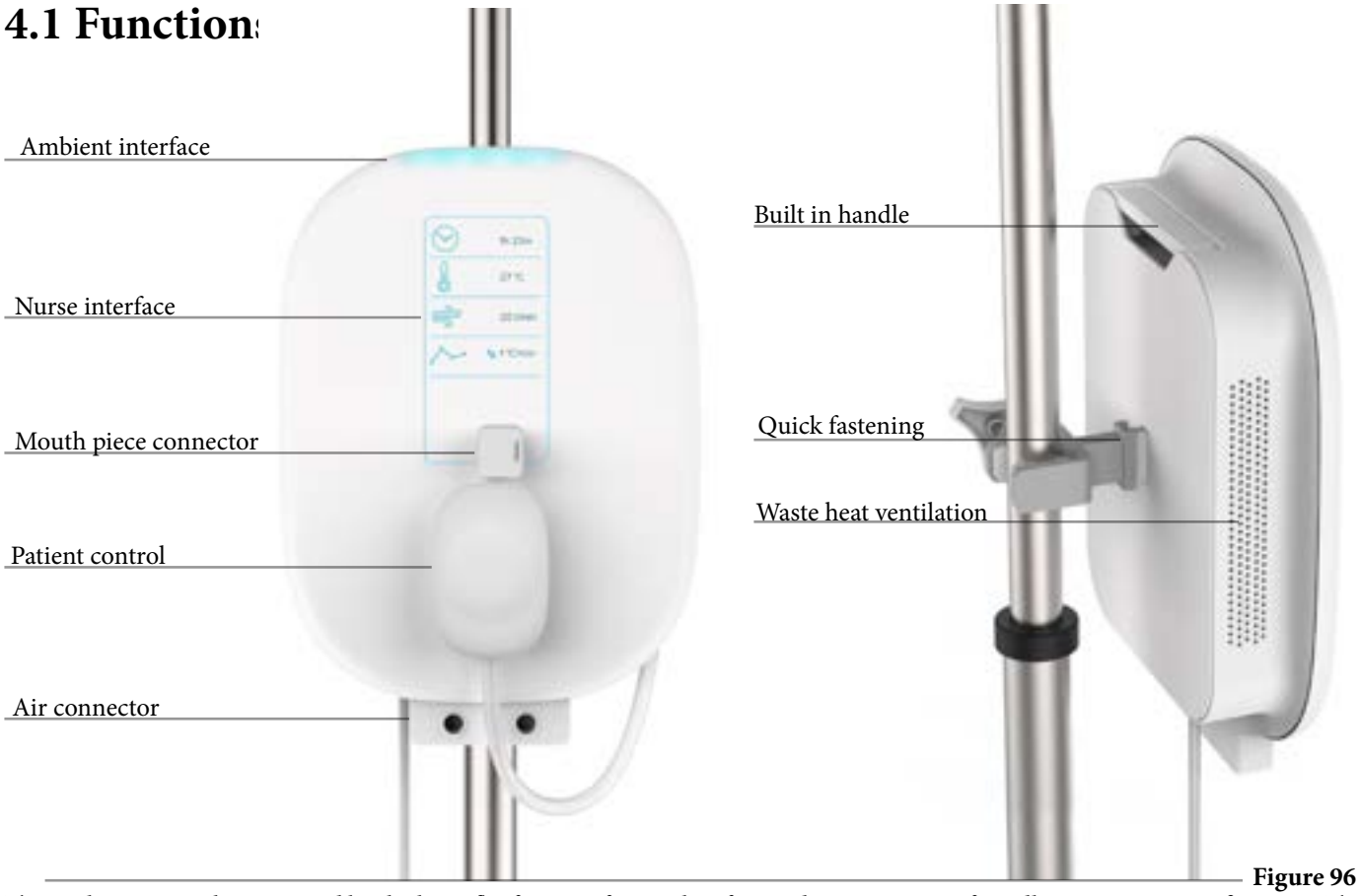


Figure 96

The cooling unit is characterized by the large flat front surface with soft roundings to create a friendly appearance, see figure 96. The main shape points towards the front where all parts that the user interacts with are placed to communicate simplicity and structure. When nurse interface is not used it automatically turns off and the screen disappears until the it is touched again, making the cooling unit less visible during the long treatments. The intention of this function is to eliminate annoying constant lights when the unit is used in a dark room. The ambient interface on the top of the unit is also back-lit through the outer plastic shell and communicates the cooling status with simple pulses of colored light. Due to the rounded shape the ambient interface light is visible from all angles, except from straight behind. Air holes for the fans cooling the thermoelectric elements are located on the side.



Figure 97

To ease handling of the cooling unit between patient room and storage, an area for winding the hose was designed as a part of the main shape, figure 97. The hose winding is held in place by the handle on the top and the electrical wire on the bottom. To preserve the minimalistic and clean appearance of the front, the handle was embedded into the casing on the back of the cooling unit. A cut out in the back of the casing directs the upwards motion required to lift up the handle. The indent on the front clearly indicates the position for the patient control unit when not in use. The control attach to the docking area with magnetic force and is wireless charged through induction.



Figure 98

The shape of the patient control unit follows the same form language as the cooling unit ¨featuring a soft, friendly and minimalistic appearance, figure 98. The white translucent silicon rubber material gives the control a rich haptic quality and enables it to transform from a white unobtrusive object to the vivid and colorful brain of the device when the button is pushed. The dynamic quality creates a sense of soulification while making sure the distraction is only present when called for.





Figure 99

While the LED diodes hidden under the silicon rubber dial is visible in full indoor lighting, figure 99. The full effect of the light is really coming forward when used in a darker room creating a more immersive and calming experience though fascination.

4.2 User scenario



1. Preparation
Introduce patient to the mouthpiece and control unit in the preparation room, let them play with it to create familiarity and de-dramatize the treatment. This should could take place before the treatment in conjunction to preparation for chemotherapy.



2. Install device
Before the chemotherapy is about to begin the nurse takes out the cooling unit from storage, carries the device to the treatment room and attach it to the iv pole. Air hoses, patient control unit and mouthpiece connector are attached to the device to form a simple to carry package.



5. Connect mouthpiece
The patient control unit and mouthpiece connector is detached from the loading dock on the front. A disposable mouthpiece in patient appropriate size is connected to mouthpiece connector after which the control and mouthpiece is given to the patient.



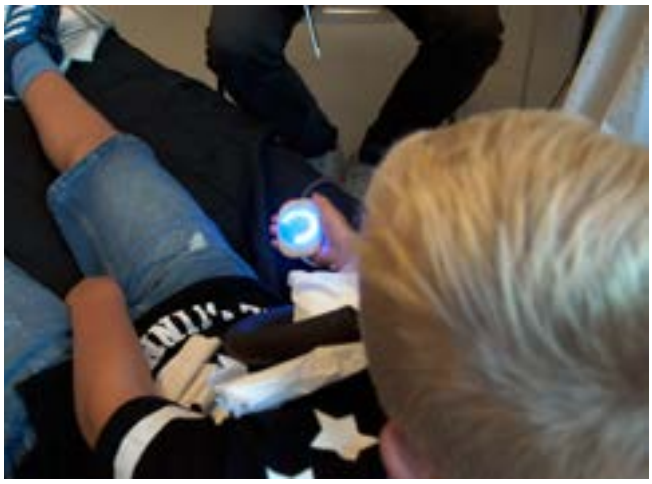
6. Start chemotherapy
Nurse starts the chemotherapy, this is a moment that requires the full attention of the nurse. Meanwhile the patient and parent discover the functions of the patient control unit in the background.



3. Connect air
Once the cooling unit is attached to the IV pole. The air inlet hose is unwound and connected to the cooling unit and the medical air outlet in the treatment room.



4. Enter treatment settings
When connecting the unit to a power supply, a start button appears on the nurse interface. The nurse press start and proceed to enter total cooling time and target temperature on the display.



7. Child initiate treatment
When the patient has gotten accustomed to the device they are asked to initiate the treatment by gradually turning on the air using the control. After the cooling has been initiated the cooling unit will automatically start the gradual cooling by slowly decreasing the air temperature and increase airflow until target temperature is reached.



8. Observe treatment
The treatment continues throughout the whole duration of the chemotherapy. Parents can supervise treatment status through the ambient interface on top of the cooling unit. Every 15 minutes a pulse of light indicates the treatment status. Orange light means the cooling is not sufficient which gives the parent an opportunity to try and motivate their child to cool without pausing too much. Blue light indicates sufficient cooling and the parent get to complement their child for doing a good job. If the cooling does not work properly a red light will pulsate constantly to urge parents to call for medical staff.

Figure 100-107. *User scenario collage*

5. Validation

After completing the final design and building prototypes the air cooling treatment was tested. Two tests were carried out, one cooling efficiency test to determine if it is technically feasible to replace ice cooling with air cooling and one to assess the difference in user experience between air and ice cooling.

5.1 Cooling efficiency test

A final test to validate the feasibility of the air cooling principle was performed. The main topic of interest for this test was to see which mouth temperatures that could be achieved with air cooling, and to compare those with ice cooling. It was also of interest to assess how demanding the air cooling is perceived during longer periods of cooling than what had been tested before .

Participants
Due to time limitations and difficulties in recruiting participants the test was only performed by one person who is also one of the authors of this thesis.

Apparatus
A pressurized tank with medical air was connected to a pipe heat exchanger, see figure 108, which in turn was cooled by a small vapor compression refrigeration system, see figure 109. The cooling system was powered by a small compressor with a cooling capacity of 25 Watts at an evaporation temperature of -17°C . The condenser of the system was cooled by a fan connected to an air heat exchanger. The refrigeration system was powered by a battery, see figure 110. At the end of the cold air hose a 3D-printed mouthpiece was connected, see figure 111.

Stimuli
Cold air at a temperature of -3,5 to -5,2 °C where injected into the mouth through the mouthpiece.

Procedure
The cooling system was allowed to run until the air temperature had reached -3,5°C after which the test began. The airflow was set to 58,8 liters per minute and was held constant throughout the test. The duration of the test was, due to lack of compressed air, limited to 34 minutes. The test person was sitting down throughout the whole period of the test see figure 112. Temperature measurements where taken at six points inside the mouth, see appendix E, before, immediately after and three minutes after the cooling.

Results
Temperature measurements immediately after 36 minutes of cooling showed that the mean temperature inside the mouth was lowered to an average of 24,7 °C which can be compared to a study on ice treatment that showed a mean temperature of 23,8 °C after 60 minutes of cooling. This strongly indicates the feasibility of substituting ice cooling with air cooling. It should however be noted that no ideal target temperature for cryotherapy has yet been established. The temperature after air cooling was relatively uneven ranging from 19,4°C on right side of the tongue to 29°C on the floor of mouth i.e. under the tongue which is similar to ice cooling, see appendix D for full results. However, more even temperatures where achieved on left and right cheek as well as left and right tongue. Improver temperature uniformity might be achieved by adjusting the air channels in the mouthpiece.

Measurements three minutes after ended cooling shows a quick rewarming of the mouth with a mean temperature difference of 5 °C comparing to measurements directly after cooling. This could indicate that the cooling is shallow and has not spread deep down in the tissue. Weather this effect is positive or negative cannot be said and due to the lack of research on ice cooling there are no results to compare with regarding this matter.

The cooling sensation was perceived as moderate during the first ten minutes of the test after which it was perceived as low. Thus, it seemed like the air temperature could have been significantly lowered, had the cooling unit been able to produce lower temperature, after the initial phase without reaching nearly the same discomfort as with ice cooling.



Figure 108. *Pipe heat exchanger. Air flows through the large opening and cooling medium in the small red opening.*



Figure 109. *Vapor compression cooler with insulated pipe heat exchanger on top.*



Figure 110. *Complete functional cooling unit prototype. Medical air tank is seen on the left, battery at the bottom and vapor compression cooler on top.*



Figure 111. *3D-printed mouthpiece.*



Figure 112. *Participant during cooling efficiency test.*

5.2 User experience test

A user experience test was conducted to asses how the air cooling device is perceived in comparison to a standard ice treatment. Although mouth temperatures where measured they were not expected to be representative of an actual cryotherapy treatment given the drastically reduced treatment time. However, mouth temperatures still needs to be taken into consideration when comparing the two treatment. The main focus of the test was on interviewing the participants after completing both tests to receive their subjective experience of the two treatments. Open questions were used in the beginning of the interviews in older to not only receive feedback on specific parts of the design but also see what the made the biggest impression on the participants.

Participants

Nine healthy children in the ages between four to twelve years participated in the test. With a mean age of 8 the test group represents the lower end of the age span of intended users (4-18). Conclusions of how teenagers would respond to the air treatment can therefore not be drawn, however it was seen as beneficial to have the lower age groups represented as they are thought to be more likely to perceive the unfamiliar air cooling as more intimidating then ice. It was also hypothesized that evaluating the patient control unit with the youngest part of the user group was a better way of determining if the interface is understood by the whole user group than to test it on older children.

Stimuli

Participants were subjected to short versions of oral cryotherapy both through ice and air cooling. The ice test was performed with crushed non-flavored ice which was self fed by the test subjects themselves with a spoon, see figure 113. In the air cooling test, the same setup as described in the cooling efficiency test presented in previous section was used, see figure 114. The air temperature during the test was measured at -3°C. A model of the final design for the cooling unit was placed close to the function prototype of the cooling unit which was hidden behind a piece of cloth, see figure 115. The arrangement was made so that it would seem like the model for the cooling device was the one actually providing the cold air. A tank of compressed medical air was also hidden and used as air supply. The air flow was manually regulated by the test leader through opening and closing a valve on the tank. In order to simulate that the air could be controlled through the patient control, the test leader observed the test subjects input to the control and tried to regulate that air accordingly. The patient control unit prototype was programmed to function identically to the previously described final design, except from being connected to an Arduino micro processor by an electrical wire, see figure 116.

Procedure

The test was held in a large studio at The royal institute of technology in Stockholm. The studio was divided into three parts using screen walls, two of the enclosed spaces were dedicated to the two tests and the third was made into a play area where children waiting to participate in a test could entertain themselves.



Figure 113. *Participants were asked to feed themselves with crushed ice using a spoon. Entertainment during the test was offered in the form a film shown on computer.*

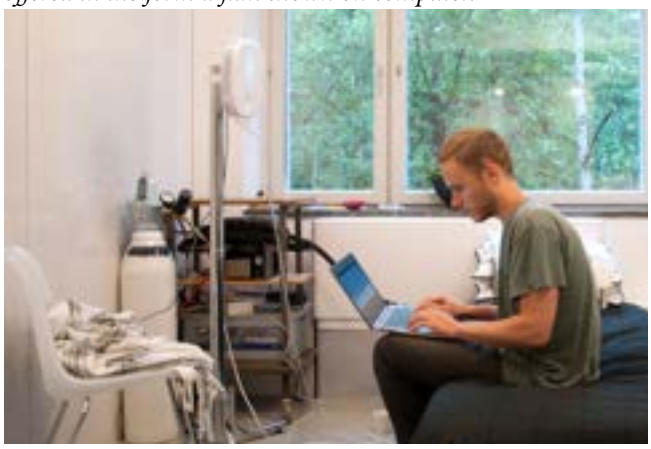


Figure 114. *The apparatus used in the air cooling test can be seen in the far corner. The vapor compression cooler with battery was covered with a cloth during the test.*

After an introduction to the test setup, participants where assigned a time slot for each of the tests. This was done to ensure they had enough time to let the mouth temperature go back to normal before starting a second test. Each of the participants where asked to complete 10 minutes of cooling per test where they cool as much as they feel they can endure. The air cooling test was proceeded by a short introduction to the patient control unit, see figure 117. Due to unexpected lack of time from the participants the cooling time had to be reduced to 5 minutes per test for some of the participants, see appendix D.

The temperature of right and left cheek was measure after completion of each test. The test subjects were encouraged to report any side effect such as nausea, headache or shooting pain in the teeth. After having completed the two tests the subjects where asked to compare the two treatments, state which one they preferred and motivate why. In interviews were the interviewees opinion on the patient control unit was not spontaneously brought up it was specifically asked for. As a way of determining if participants had understood the interface of the control they were asked to recall the functions and explain how they worked.



Figure 115. *Test leader introducing the patient control to a participant, functional cooling unit is hidden under the cloth in the background.*

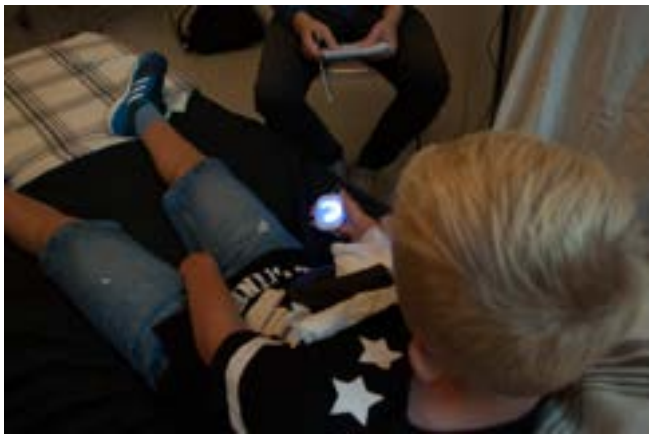


Figure 116. *Participant using the control to adjust airflow during air cooling test.*

Result

The results from the temperature measurements show that the ice treatment resulted in an average of 4-6 °C lower mouth temperature. Although this was expected due to the short cooling time, it should be kept in mind when assessing the results of the subjective comparison. It should also be pointed out that four of the tests for each cooling method, see appendix E, was reduced to five minutes which might effect the comparison. It was also found to be difficult to interview some of the participants who were reluctant to answer some of the interview questions. This was thought to be explained by either shyness or restlessness.

The interview results showed that six out of nine participants preferred the air cooling. The main disadvantage of ice cooling was reported to be that it was hard to chew the ice and uncomfortable with the melting water. Ice cooling was perceived as more active compared to the more passive air cooling. Out of those who preferred ice cooling, one participant referred to it as less intimidating compared to air cooling. Another participant stated that the air flow was mistakingly too high in the beginning as a reason for preferring ice cooling.

A commonly mentioned problem with the air cooler was excessive salivation. The problem is thought be explained by air blowing on the salivation glands causing excessive saliva production. If the head is bent forward during cooling the saliva risk to be blown out in the exhaust air channel of the mouthpiece. It is however unclear whether this will be a problem for the intended user group since mouth dryness is



Figure 117. *Participant getting used to the patient control before test.*

"It felt better when I could control the cooling... I did not use the control to adjust much, but it is nice to know that you can if you want to." - 11 year old child

a common side effect from chemotherapy which might even lead to the salivation problem in this test to become a benefit of the cooling method.

The perhaps most important result of the test was that five out of nine participants spontaneously mentioned the ability to control the air when asked to explain why they preferred the air cooling. Two participants referred to the patient control as "good to have" when specifically asked about it. This suggest that the patient control has an important effect on creating a sense of control for the patients. Only one of the participants claimed to be intimidated in the beginning by the fact that the control over the treatment was given to them, however it was also stated that he still preferred to control the air himself rather than having an adult controlling it.

The hypothesis that the sense of control is more important than actually control the device was confirmed by one participant who motivated his preference for air cooling over ice with that it felt better when he could control the treatment. When asked if he used the control to adjust the air during the test he replied that he did not, but that it still felt good knowing he could.

The complexity of the interface was proven to be well adjusted given that all of the participants who agreed to take part in the individual interview following the two tests were able to recall and explain all of the four functions on the interface. The general opinion of those interviewed was that they were able to quickly learn how the interface worked.

When asked none of the participants had any particular thoughts on the cooling unit model but claimed to have their focus on the control unit instead. Through observations is seemed that all participants where interested by the control and the light coming from it. Suggesting the it had a distracting effect. The youngest participant (four year old) expressed the most fascination for the control claiming it was "superduper cool".

6. Discussion

Designing an entirely new type of product proved to be very time consuming. Functions essential for the end product otherwise validated through an existing product had to be designed from scratch.

It was therefore sometimes difficult to do research at the hospital department, interviews and observations were often restricted due to a limited amount of participating patients and occupied hospital staff working under tight schedules. Because the majority of tests could not be performed at the hospital, this ways a major limiting factor since the context and conditions in which the designed product will function in are very hard to replicate elsewhere.

The goal of the project was to increase the treatment acceptance as much as possible. Although the first phase was focused on minimizing the discomfort by finding a better means of cooling than ice, It was realized in an early stage that acceptance through comfort is limited. This is due to the fact that cooling the mouth to a low temperature for several hours will always be uncomfortable. Therefore, psychological means of increasing acceptance was also included early in the process.

The wide user group ranging between four and eighteen years old proved to be challenging when it comes to designing something that grabs the children's attention. By making the level of attention optional and based on fascination rather than entertainment, the concept was designed to fit users through the whole age span.

In the final test with healthy children, see chapter 5.0 *Validation*, the concept of a patient control was very well received. Some participants motivated their preference for the air cooling with the fact that they could control the cooling. Suggesting that the design has a real impact on the sense of control. Some participants also referred to the light effects of the patient control unit as cool which indicates that the ambition to fascinate children with the light holds some merit.

The test also showed that all participants, who agreed on being interviewed, could both recall and explain all functions of the interface. This suggest that the level of complexity of the interface is well adjusted to the abilities of a large part of the user group. A lack of participants from the youngest in the user group, five to four years of age, prevents any conclusions to be drawn about how well they would understand the interface.

Designing a product for children was very stimulating but also challenging. A "prototype fast fail fast" methodology in which crude mock-ups is frequently tested and evaluated by the intended end users could not be applied in this project, both due to restricted access to users, but mainly due to the fact that the user group is children. It was discovered early in the project that especially young kids find it difficult to imagine something in a test that is not there. Even when building a functioning light interface with a push button to test different interface positions the test subject had a hard time understanding the context of the test because there was no cooling unit to control. This motivated the time spent on building working prototypes.

Regarding the cooling technique the final test showed temperatures similar to those measured during ice treatment. Thus the main requirement set for cooling capacity can be seen as fulfilled. It should be pointed out that the test was not in a clinical setting and should therefore be regarded as a reason for continued studies rather than scientific proof. It does however suggest that a cooling unit with only 25 watts of cooling capacity is enough to cool a mouth to temperatures comparable to those with ice cooling. This strongly supports the feasibility of the compact cooling unit design and show that it is plausible that the cooling unit could be of such size that it is manageable to mount it on an IV-pole. This is further strengthened by the fact that a far from optimal heat exchanger was used to cool the air.

A large challenge regarding cooling was to achieve uniform cooling throughout the oral cavity. The final air cooling test shows relatively even mouth temperatures when comparing measurements on left and right side. However the tongue and cheeks where cooled to a considerably lower temperature than the floor of mouth and soft palate. Although more tests are needed d to make a good comparison to the ice treatment regarding this matter, the test results suggest that air cooling hold the potential for a more uniform cooling. Further testing and detailed design of the mouthpiece and in particular the size and direction of the air inlet channels as described in following chapter is considered necessary.

The need for further testing during longer cooling periods in the correct hospital environment could not be understated. It is needed to determine the effect of gradual cooling, parents ambient interface, integration in nurse routines and the devices influence on existing distractions. The test carried out in this project does neither contradict, nor prove the presented hypothesis on how the design would perform on these subjects.

7. Recommended future work

The dimensions of the cooling unit was approximated based on the size of products with similar functions, since a decision of specific components was not included in the scope of the project. Choice of specific internal components should be further investigated and researched.

Although mobility regarded as an important by both medical staff and parents the device has not been specifically designed for this. The technical complexity involved in judging how the device should best be designed to allow maximum mobility was deemed too time consuming to be a part of this project. Instead the device was designed to be as simple, small and lightweight as possible to offer the possibility of building a mobile solution with external components is requested. Since the cooling unit is attached onto the IV-pole it can easily be moved around with chemotherapy equipment during the treatment. A mobile setup can easily be created by attaching a tank with compressed air and an external battery to the IV-pole.

The recommended next step in the process of getting the product ready for the market is to make necessary adjustments to the vapor compression cooling prototype developed in this project to make it sufficient for clinical trials. During the process of clinical trials the evaporator, the heat exchanger between the cool side of the refrigeration device and the air, should be optimized. Subsequently, thermoelectric elements should be dimensioned to provide enough cooling capacity. The required capacity, including safety margin, can be estimated to 50 watts by comparison to the tests with the vapor compression cooler. Special attention should be brought to the dimensioning of heat sinks and fans too make sure the large amount of waste heat produced by the thermoelectric elements is properly discarded.

Furthermore it is recommended to release a first version where the air cooling is as simple and cheap as possible to reach the market fast. In this regard the final design presented in this report is considered a good design for a first generation of devices as the complexity is minimized by utilizing wall outlets for both air and electricity. Once the first generation of devices are built they should be tested extensively to determine how large the need for mobility really is and is the market is ready to pay for the increased cost of a more mobile product.

It mobility is proven to be of high priority, two changes should be considered. Firstly the refrigeration device could be changed from thermoelectric elements to a vapor compression system. This would increase the weight of the cooling unit itself, but at the same time reduce the weight of the battery. Specifications on required time of use in mobile mood together with further calculations and tests are needed to demain if this change is beneficial to the end user. The second change that could be made to improve mobility is to have a built in dehumidifier which allows the device to use ambient room temperature air directly instead of compressed dry air. Once again this will add to the size and weigh of the cooling unit but also eliminate the need for an air tank to provide mobility. Both these changes has to be further evaluated to ensure that they actually improve mobility and the overall user experience before being implemented. Regarding the technical feasibility the Zimmer cryo 6, presented in chapter 2.4 *State of the art,* can work as a reference. It should be noted that the Zimmer device delivers 1000 liter per minute at -30°C while the final cooling test indicates that 60 liters per minute of -5°C air was sufficient to reach target temperature, meaning the mouth cooling device should be able to be considerably smaller that the Zimmer device even with built in dehumidifier. The final cooling test also showed that a 7,2 ampere-hour battery was sufficient for 50 minutes of cooling with a vapor compression system.

The mouthpiece should be further developed through an iterative process of multiple tests. The air inlet channels may be adjusted to achieve a more uniform and comfortable cooling. The currently most pressing problem to solve regards the exhaust air channel. The current design allows saliva to get into the exhaust air channel resulting in it dripping from the mouthpiece onto the patient. Furthermore the exhaust channel should be resigned in such way that i can not be accidentally blocked by the tough which is currently the case, causing momentarily inflation of the mouth which can be booth uncomfortable and frightening for the patient.

The design of the mouthpiece should include a thermometer to make it possible to keep track of the current treatment and make sure that the mucosa reaches the desired temperature. Ideal case would be to measure directly on the mucosa but if it is not possible or too uncomfortable other ways of measuring the temperature should be investigated. For instance the exhaust air could be measured to obtain a rough estimation of the mucosal temperature. The accuracy of the chosen method should be investigated and validated.

It is recommended to make more tests especially on the real end user during chemotherapy to demain if it is needed to add moisture the dry air cold air. Salivation and condensed water inside the mouth has to be considered so that moist air does not require the patient to swallow or spit too often. If air moisturizer is incorporated it should be located at the end of the hose close to, or inside the mouthpiece to prevent condense water to gather inside the hose. Either passive moisturizing by letting the dry air blow over a small water container could be used. Alternatively, active moisturizing where a small vibrating piezoelectric element place in the water crates a fine mist that is dispersed in the cold air.

It is recommended to investigate the need for insulation of the hose between the cooling unit and mouth piece to maintain a constant air temperature and minimize the required cooling capacity. Isolation could also be needed to avoid the discomfort that might occur from the cold hose touching the skin of the patient.

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9. Appendices

A - Observation guidelines and questions for user test at the hospital I
B - Task sheet with explanatory pictures for user tests at the hospital II
C - Code for prototype III
D - User experience test V
E - Cooling efficiency test VI

Appendix A - Observation guidelines and questions for user tests at the hospital

The appendix contains observation guidelines and questions used during user tests at the hospital. A large part of the data from the test was gathered by observing the users interaction with the mock-ups and each other. The list below outlines the major topics of observation.

Observations

In general
Does the patient control unit contribute to a feeling of control?
Does the interface capture the interest by the child and thereby the ability to distract? from the treatment in a positive way?
What is the dynamics between the different users; child, parents and nurse?
How well does the child understand the depiction of time the through the LED ring?

For each concept
How visible is the interface for the different user groups?
How accessible is the interface for the patient?
Does the interface imply any restrictions to the patient?

Questions

Which concept was your favorite and why?
How does the parent feel about their child controlling a treatment?
What kind of time indicators of the treatment are used during chemotherapy?
Does the child often ask how much time there is left of the treatment?
How does the nurse provide feedback on the patient's performance concerning the treatment?

Appendix B - Task sheet with explanatory pictures for user tests at the hospital

The appendix contains the task sheet, see figure 1, used during user tests at Uppsala University Hospital. The sheet contains different variations of color circles used to test the round LED patient control interface. The participants task was to produce each of the colour dot combinations as way of creating a meaningful interaction with the interface in place of controlling a functioning cooling unit. Figure 2 shows a child friendly illustration of the device which was used to explain how it would work.

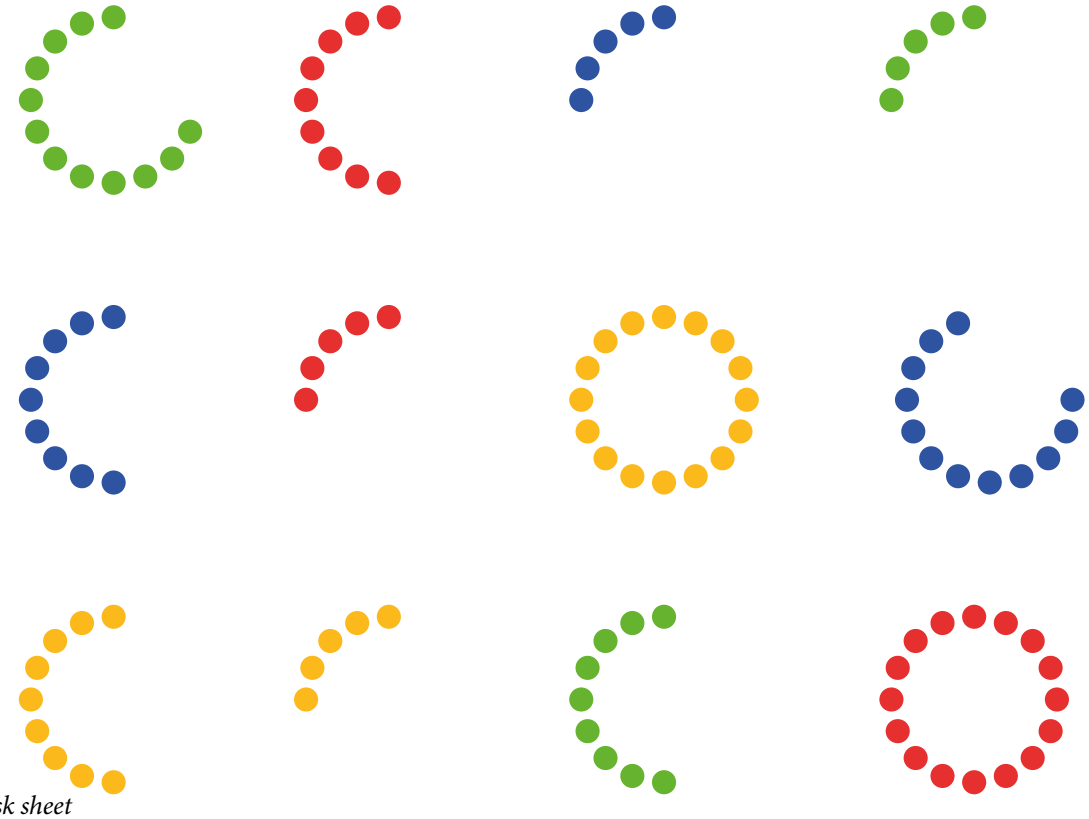


Figure 1. *Task sheet*

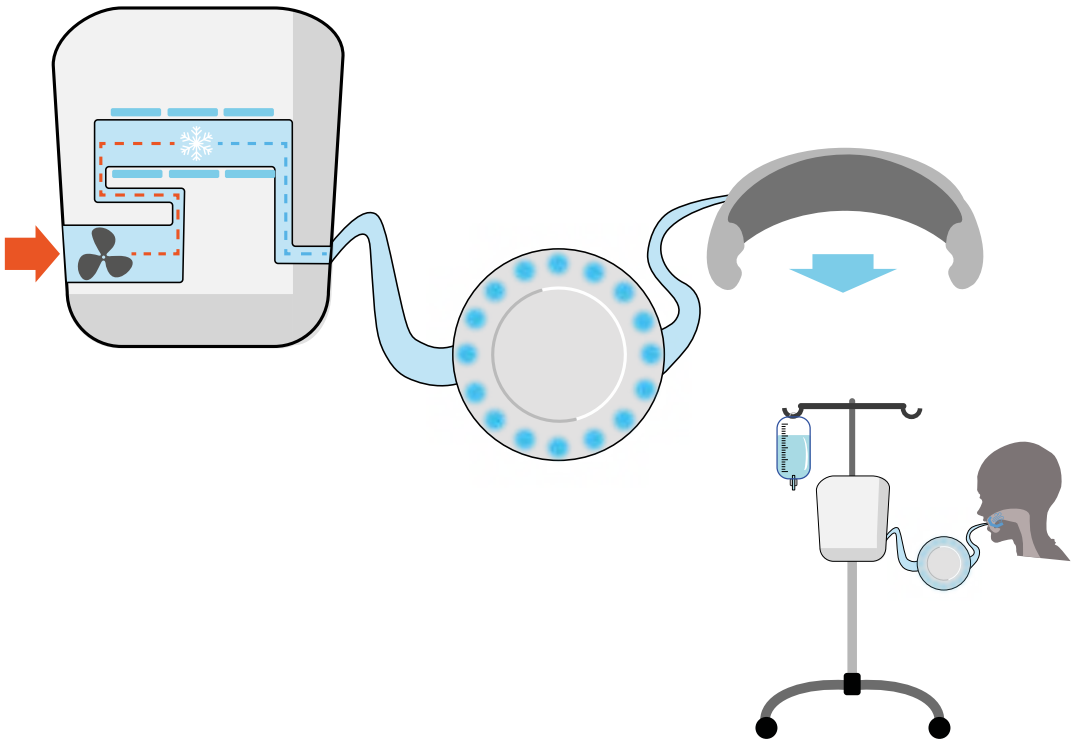


Figure 2. *Child friendly illustration of the whole cooling device.*

Appendix C - Code for prototype

Presented below is the code used to control the interfaces of the cooling and control unit prototypes. The code is written in the programming language C++ and executed with an Arduino Uno microprocessor.

```
#include <Adafruit_NeoPixel.h>
#include <math.h>
#define PIN_kontroll 11
#define PIN_ke_topp 8
#define PIN_ke_display 6

volatile boolean TurnDetected;
volatile boolean up;

//Rotary Encoder variables
const int PinCLK=2;
const int PinDT=4;
const int PinSW=7;
static long virtualPosition=0;
int ScrollValue = 0;
bool timeMenu = false;
float timeMenuTime = 0;

//Menu navigation variables
bool Idle=true;
bool StartMenu=false;
bool PausMenu=false;
bool TimeMenu=false;
bool BlowMenu=false;
bool TempMenu=false;
long int StartTime = 0;
int temp = 0;
volatile int BlowValue = 0;

Adafruit_NeoPixel leds_kontroll = Adafruit_NeoPixel(20, PIN_
kontroll, NEO_GRB + NEO_KHZ800);

Adafruit_NeoPixel leds_topp = Adafruit_NeoPixel(12, PIN_ke_
topp, NEO_GRB + NEO_KHZ800);

Adafruit_NeoPixel leds_display = Adafruit_NeoPixel(24, PIN_ke_
display, NEO_GRB + NEO_KHZ800);

void setup () {
 leds_kontroll.begin();
 leds_kontroll.show();

 leds_topp.begin();
 leds_topp.show();

 leds_display.begin();
 leds_display.show();

 pinMode(PinCLK,INPUT);
 pinMode(PinDT,INPUT);
 pinMode(PinSW,INPUT);
 attachInterrupt (0,isr,FALLING);
 Serial.begin (9600);
 Serial.println("Start");
}
void loop () {
 run_display();
 run_topp();
 Snooze();
 if (!(digitalRead(PinSW))) {
  StartTime = 0;
  delay(300);
  ChangeMenu();
  virtualPosition=0;
  Serial.print ("Reset = ");
 }

 if (TurnDetected) {
  StartTime = 0;
  TurnOffLeds();
  if (up)
   if (virtualPosition >= 15 && (!StartMenu)) {
    virtualPosition = 0;
   }
   else if ((virtualPosition >=3) && (StartMenu)) {
    virtualPosition = 0;
    TurnOffLeds();
   }
   else {
    virtualPosition++;
   }
   else
    if (virtualPosition <= 0 && (!StartMenu)) {
    virtualPosition = 15;
   }
   else if ((virtualPosition <=0) && (StartMenu)) {
    virtualPosition = 3;
    TurnOffLeds();
   }
   else {
    virtualPosition--;
   }

   TurnDetected = false;
 }
 // Menu Mode
 if (Idle){
  TurnOffLeds();
 }

 else if (StartMenu){
  leds_kontroll.setPixelColor(16 + virtualPosition, 255, 255, 255);
  leds_kontroll.show();
 }

 else if (PausMenu){
  int u;
  for (u = 0; u < 500; u+= 2) {
   delay(1);
   int y;
   for (y = 0; y < 16; y++) {
    if (!digitalRead(PinSW)) {
     return;
    }
    if (u > 250) {
     leds_kontroll.setPixelColor(y, 500 - u, (500 - u)/3, 0);
     leds_kontroll.show();
    }
    else if (u < 251){
     leds_kontroll.setPixelColor(y, u, u/3, 0);
     leds_kontroll.show();
    }
   }
  }
 }

 else if (TimeMenu){
  int TimePassed = 5;
  int u;
  for (u = 0; u < 500; u+= 2) {
   int y;
   for (y = 0; y < 16; y++) {
```

```
      if (!digitalRead(PinSW)) {
        return;
      }
      if (y < TimePassed){
        leds_kontroll.setPixelColor(y, 20, 20, 60);
        leds_kontroll.show();
      }
      else if (y > TimePassed) {
        leds_kontroll.setPixelColor(y, 60, 20, 20);
        leds_kontroll.show();
      }
      else if (y == TimePassed)
      if (u > 250) {
        leds_kontroll.setPixelColor(TimePassed, 0, 0, 500 - u);
        leds_kontroll.show();
      }
      else if (u < 251){
        leds_kontroll.setPixelColor(TimePassed, 0, 0, u);
        leds_kontroll.show();
      }
    }
  }

  else if (BlowMenu){
    int i;
    for (i = 0; i < BlowValue; i++){
      leds_kontroll.setPixelColor(i, 0, 0, 40);
      leds_kontroll.show();
    }
    if (virtualPosition != 15){
      for (i = BlowValue; i < 16; i++){
      leds_kontroll.setPixelColor(i, 0, 0, 0);
      leds_kontroll.show();
      }
    }
    BlowValue = virtualPosition;
  }

  else if (TempMenu){
    if (temp == 0){
      temp = random(15);
    }
    int i;
    for (i = 0; i < temp; i++) {
      leds_kontroll.setPixelColor(i, 150 - i*10 , 50 - i*1/3, i*10);
      leds_kontroll.show();
    }
  }
}

void ChangeMenu() {

  if(Idle){
    TurnOffLeds();
    Idle = false;
    StartMenu = true;
  }
  else if ((StartMenu) && (virtualPosition == 0)){
    TurnOffLeds();
    StartMenu = false;
    PausMenu = true;
  }
  else if ((StartMenu) && (virtualPosition == 1)){
    TurnOffLeds();
    StartMenu = false;
    TimeMenu = true;
  }
  else if ((StartMenu) && (virtualPosition == 2)){
    TurnOffLeds();
    StartMenu = false;
    BlowMenu = true;
  }
  else if ((StartMenu) && (virtualPosition == 3)){
    TurnOffLeds();
    StartMenu = false;
    TempMenu = true;
  }
  else {
    TurnOffLeds();
    StartMenu = true;
    PausMenu = false;
    TimeMenu = false;
    BlowMenu = false;
    TempMenu = false;
    Idle = false;
    temp = 0;
  }
}

void TurnOffLeds() {
  int i;
  for (i = 0; i < 20; i++){
    leds_kontroll.setPixelColor(i, 0, 0, 0);
    leds_kontroll.show();
  }
}

void Snooze() {
 if (StartTime == 0) {
  StartTime = millis();
 }

 if (millis() - StartTime > 20 * 1000) {
  Idle = true;
  StartMenu = false;
  PausMenu = false;
  TimeMenu = false;
  BlowMenu = false;
  TempMenu = false;
 }
}

void run_display() {
  int i;
  for (i = 0; i < 6; i++){
    int red = 40;
    int green = i*20 + 90;
    int blue = 255-i*40;

    leds_display.setPixelColor(i,    red, green, blue);
    leds_display.setPixelColor(12+i, red, green, blue);
    leds_display.setPixelColor(11-i, red, green, blue);
    leds_display.setPixelColor(23-i, red, green, blue);
    leds_display.show();
  }
}

void run_topp() {
 int i;
 for (i = 4; i < 8; i++){
  leds_topp.setPixelColor(i, 10, 150, 150);
  leds_topp.show();
 }
}
void isr ()  {
  if (digitalRead(PinCLK))
    up = digitalRead(PinDT);
  else
    up = !digitalRead(PinDT);
  TurnDetected = true;
}
```

Appendix D - Cooling efficiency test

Presented below are the results from the cooling efficiency tests. The test consisted of cooling the mouth with air for 34 minutes. The air temperatures during the test is presented in table 1 below. The mouth temperatures measured before, immediately after and three minutes and after the test is presented in table 2 below. Table 3 shows data obtained in a another study where healthy test subject were subjected to 60 minutes of cooling with crushed ice.

Time	Test start	After 21 minutes	After 34 minutes
Air temperature	-3,5	-5,2	-4,1

Table 1: The table shows air temperature measured during the test.

	Buccal mucosa Right	**Buccal mucosa left**	**Right tongue**	**Left tongue**	**Floor of mouth**	**Soft palate**	**Mean**
Start °C	36,0	36,2	36,1	35,5	36,0	34,8	35.8
36 min °C (test end)	24,0	24,0	19,4	23	29	28,5	24,7
36+3min °C	28,4	28,5	28,3	28,2	32,5	32,2	29,7

Table 2: The table shows mouth temperature measured at six locations before, immediately after and three minutes after the test. Buccal mucosa refers to the inside of the cheek.

	Before experiment session	After experiment session	Temperature difference	*p* value
All surfaces	35.7 (SD 0.9)	23.8 (SD 3.0)	12.9	<.002
Buccal mucosa right	36.3 (SD 0.7)(35–37)	23.3 (SD 4.5)(14–8)	13.6	<.002
Buccal mucosa left	36.6 (SD 0.5)(36–37)	22.8 (SD 5.9)(14–32)	14.7	<.002
Labial mucosa upper	35.3 (SD 1.7)(32–37)	24.1 (SD 3.4)(17–31)	11.1	<.002
Labial mucosa lower	36.1 (SD 1.4)(34–38)	23.5 (SD 2.6)(12–18)	13.0	<.002
Tongue dorsal anterior	35.2 (SD 1.4)(33–38)	21.3 (SD 6.0)(9–29)	13.6	<.002
Tongue dorsal posterior	35.8 (SD 1.0)(34–37)	25.8 (SD 3.5)(17–31)	8.5	<.002
Tongue dorsal ventral	35.7 (SD 1.4)(32–37)	26.8 (SD 3.0)(22–31)	10.6	<.002
Hard palate	34.9 (SD 1.1)(33–36)	22.8 (SD 3.9)(15–27)	12.2	<.002

Table 3 : Temperature measured before and after 60 minutes of cooling with ice.
Source: Svanberg, A., Öhrn, K., Broström, H. and Birgegård, G. (2012). The effect of cryotherapy on oral mucosa: a study in healthy volunteers. Medical Oncology, 29(5), pp.3587-3591.

Appendix E - User experience test

Presented below are the results from the user experience test. The test consisted of cooling the mouth with ice and air for ten minutes each. Participants where then asked to compare the two treatments as well as explaining the interface function to asses their understanding of the same. All of the 10 participants are presented in the table 1 on table 2 below.

Test subject	no1	no2	no3	no4	no5	no6	no7	no8*	no 9*	**Mean**
Age	11	9	7	12	7	9	4	7	6	8
Ice treatment	*				*		Aborted	*	*	
Temp(°C) Right cheek	27	35,2	31,3	30,1	30,8	32,2	-	32,2	22,3	30,11
Left cheek	31,6	36	31,6	30,4	33,6	32,5	-	31,1	28,2	28,33
Air cooling			*			*	Aborted after 3min	*	*	
Temp(°C) Right cheek	34,8	32	36	32	34,9	No temp. measurement obtained	-	No temp. measurement obtained	No temp. measurement obtained	34
Left cheek	35,5	35	36,3	32,1	35,6	No temp. measurement obtained	-	No temp. measurement obtained	No temp. measurement obtained	34,9

Table 1: Measured mouth-temperature. On three of the air cooling tests the temperature could not be measured due to thermometer malfunction. * Performed test for 5 minutes due to lack of time.

Test person	**Subjective comparison**	**Patient interface understanding**
No 1	Ice treatment gives a lot of water in the mouth but good when cooling short periods. Prefers air cooling for longer periods. A bit hard with salivation during air cooling. I liked that I can control the air myself, then you have the ability to adjust if it gets too cold. I understood the control well but i did not use it much, it was just nice to know the you can if you want to.	Able to recall and explain all functions
No 2	Preferred ice. The air came all at once when i began the test, but I was able to reduce the flow with the control, then it got better.	-No answer obtained
No 3	Air was less cold and a bit more comfortable. The ice was hard to chew. Uncomfortable with salivation during air treatment but it was not scary. It felt good with the control and that you can choose for yourself. I understood the functions quickly.	Able to recall and explain all functions
No 4	Prefers air colling over ice. Ice is just “troublesome”. Air cooling was less uncomfortable. With ice you have to chew and be active but with air you can just rest. It was good that you can control the air yourself, I learned the functions fast.	Able to recall and explain all functions
No 5	Preferred ice but uncomfortable with all water. The control button was somewhat hard to work.	Able to recall and explain all functions,
No 6	Prefer air cooling because it was more comfortable. The control was good, good that you can choose yourself.	Able to recall and explain all functions.
No7	Preferred air cooling. “The ice was colder. The control is “superduper cool”, it was fun to control the air.	-No answer obtained
No8	Prefers air cooling. “The control worked pretty well, I was a bit scared of doing something wrong in the beginning but I would still choose to control the air rather than having a grownup doing it for me”.	-No answer obtained
No9	Prefers ice, less intimidating. “more fun with ice”.	-No answer obtained

Table 2: The table contains a subjective comparison between ice and air cooling as well as how well participants were able to recall and explain the interface functions.